

Twenty-five structural conjectures for prime patterns, occupancy, finite logarithms, arboreal dynamics and factorization

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Abstract

Twenty-five conjectures are stated across five programmes: connected prime-pattern fields, arithmetic occupancy and local-global decoupling, finite logarithms and Kummer rank, arboreal resolution and primitive dynamical factors, and Frobenius-marked polynomial factorization. Each statement declares its probability space, separates its constructive layer from its converse layer, derives its constants from local, Galois, overlap or sieve structure or else registers the derivation as a stated programme, and carries a finite-range evidence note that reports what the computation can and cannot bear on. The organizing principle is that a conjecture should introduce a mechanism, a canonical invariant, a new scale or a functorial relationship, so that a generic limit law by itself is not admitted.

Summary sheet: the twenty-five statements

1. **Connected Hardy–Littlewood generating functional (Conjecture 1).** Every joint cumulant of centred motif fields, under the window-position sampling law, expands over the overlap poset with Möbius-connected singular-series coefficients, no coefficient occurring outside that expansion, and Gaussianity follows from a stated bound on connected cumulants of order at least three rather than being assumed.
2. **Overlap-filtration variance law (Conjecture 2).** The variance of a linear combination of motif counts expands over the two-level overlap poset of a pair, with explicit powers of $\log X$ indexed by coincidence multiplicity, annihilation of the coincidence matrices deletes that whole level, and the renormalized disjoint kernel then fixes the leading scale.
3. **Local-to-spectral covariance decomposition (Conjecture 3).** The motif covariance splits into a coincidence part indexed by the overlap poset, a renormalized connected singular-series kernel of disjoint translates, and the window pullback of the joint zero-correlation measure, with no fourth source at leading order and a crossover scale fixed by equality of masses.
4. **Spectrally conditioned extremal index (Conjecture 4).** On the window-position probability space, conditioning on the sigma-field generated by the long-wavelength component makes cluster starts Poisson, the within-cluster law is the admissible self-overlap graph, and removing the conditioning returns a Cox cluster process.
5. **Completeness of arithmetic wavelets (Conjecture 5).** Joint kernels of the overlap incidence tensors produce predicted low-variance combinations, and the converse layer asserts that every anomalously rigid coefficient sequence is asymptotically one of them.

6. **First-order arithmetic occupancy kernel (Conjecture 6).** Under the modulus-class law the least-prime survival function is $e^{-t}[1 - A(t)/\log Q + o(1/\log Q)]$ with A obtained by integrating the connected two-prime exclusion kernel.
7. **Dual additive-multiplicative spectral geometry (Conjecture 7).** The least-prime covariance operator is an additive convolution kernel plus a character-diagonal operator built from zeros of Dirichlet L -functions, with no surviving mixed term after dyadic averaging.
8. **Conditioned Cox–Poisson law for the last classes (Conjecture 8).** On the modulus-class probability space, conditioning on the finitely many non-vanishing low modes Poissonizes the terminal excess process, and unconditionally the limit is the Cox process directed by those modes.
9. **Rank-one exceptional-zero deformation (Conjecture 9).** A real zero close to 1 deforms the first-arrival field by exactly one spectral spike along the associated real character.
10. **Growing local-condition independence for odd class groups (Conjecture 10).** After the two-primary genus data are fixed, the odd class group remains Cohen–Lenstra distributed under prescribed splitting signs at all primes up to B whenever $\pi(B) = o(\log |D|)$.
11. **Kummer–Haar law for Fermat quotients (Conjecture 11).** For a saturated group of rank r the normalized Fermat-quotient homomorphism equidistributes on the relation subtorus, whose dimension is the saturated rank and not the number of displayed generators.
12. **Square-root large sieve for finite logarithms (Conjecture 12).** Nonrelation character sums of finite logarithms over dyadic prime blocks are of square-root size uniformly for character heights growing as a power of the range.
13. **Saturated-rank threshold for simultaneous Wieferich primes (Conjecture 13).** Rank one gives $\kappa(\Gamma) \log \log X$ vanishing primes and rank at least two gives finitely many, the invariance statements holding only for saturated and Kummer-compatible groups.
14. **Rank-depth Poisson law for higher strata (Conjecture 14).** Over a uniformly sampled height ensemble of generator tuples, the maximal-depth exceptional primes form a Poisson process of intensity $\kappa(r, k; p)p^{-rk} - \kappa(r, k + 1; p)p^{-r(k+1)}$, each fixed depth slice is Poisson of intensity $\kappa(r, k; p)p^{-rk}$, and the depths are joined by Haar lifts.
15. **Toric finite logarithm and codimension principle (Conjecture 15).** The first congruence quotient on any algebraic torus is Haar on the Lie algebra of the algebraic subgroup generated by the point, functorially in the torus, so the exceptional exponent is $\dim S$.
16. **Joint arboreal Chebotarev law at growing depth (Conjecture 16).** Below the entropy-conductor boundary the whole visible preimage tree is Haar–Chebotarev distributed in the finite arboreal image, jointly across levels.
17. **Sharp entropy-conductor barrier (Conjecture 17).** The upper half is a conditional proposition under a Riemann hypothesis for the preimage fields, the lower half is asserted family-averaged, and the critical-window Gaussian-multinomial profile is the conjectural clause.

18. **Poisson–Dirichlet law for primitive dynamical factors (Conjecture 18).** In a uniformly sampled parameter family with slowly growing level, the ranked logarithms of the primitive prime factors converge to PD(1), marked by first-appearance level and Frobenius data.
19. **Euler law for squarefull primitive factors (Conjecture 19).** The primitive part is squarefree with a limiting Euler-product probability whose local factors count parameters modulo p^2 , and the squarefull logarithm is tight.
20. **Geometric-arboreal criterion for large dynamical gcds (Conjecture 20).** Macroscopic orbit gcds occur if and only if the orbits lie on a periodic algebraic correspondence, the “only if” direction being the converse layer.
21. **Frobenius-marked Poisson–Dirichlet factor process (Conjecture 21).** Macroscopic prime factors of polynomial values form a PD(1) process marked by Frobenius classes with intensity size-biased by fixed-point counts, normalized to one by Burnside.
22. **Small-large independence after logarithmic mass removal (Conjecture 22).** The small-prime Kubilius field is asymptotically independent of the large-factor process once the latter is normalized by the residual mass rather than the full mass.
23. **Coloured factor superposition with local-exclusion gluing (Conjecture 23).** Conditioned on the joint local state the colour processes are independent PD(1), and unconditionally their entire interaction is an explicit Gibbs–Euler factor built from joint root-count tensors.
24. **Saddle-point entanglement law for joint smoothness (Conjecture 24).** The joint smooth-value density is the product of component probabilities times an Euler product of local root distributions tilted by the multivariate saddle-point parameters, so the entanglement factor moves with the smoothness parameters.
25. **Buchstab–Bateman–Horn sieve-to-prime bridge (Conjecture 25).** Rough-value densities follow the finite- u Buchstab law, and the conditional prime probability is the Bateman–Horn density divided by that law, with an elementary endpoint at $1 < u \leq 2$.

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1 Notation and conventions

Sampling laws

Every stochastic clause below names one of three ensembles. No clause attaches “almost surely”, a central limit theorem, a conditional law or an extremal limit to a deterministic sequence.

(W) *Window-position law*. Fix $X \geq 3$, a fixed $\varepsilon > 0$ and a window length L with $X^\varepsilon \leq L \leq X^{1-\varepsilon}$. Put $\Omega_X = [X, 2X]$ with its Borel field and the dyadic-log probability

$$\mathbb{P}_X(dx) = \frac{dx}{x \log 2}.$$

A *random window* is $[x, x + L]$ with $x \sim \mathbb{P}_X$. All fields of Programme I are random variables on $(\Omega_X, \mathbb{P}_X)$, and every sigma-field conditioned on in Conjectures 4 and 8 is a sub-sigma-field of a space declared here. Expectation, variance and covariance under $\mathbb{P}_X$ are written $\mathbb{E}_X$, Var_X , Cov_X .

(M) *Modulus-class law*. Fix $Q \geq 3$ and put

$$\Omega_Q = \{(q, a) : q \text{ prime}, Q \leq q \leq 2Q, a \in (\mathbb{Z}/q\mathbb{Z})^\times\}, \quad \mathbb{P}_Q(q, a) = \frac{1}{\pi(2Q) - \pi(Q)} \cdot \frac{1}{q-1}.$$

The modulus is uniform on the dyadic block of primes and the class is uniform given the modulus. This is the ensemble meant by “dyadic averaging” throughout Programme II.

(P) *Parameter-family laws*. In Programme III, Conjecture 14 samples an ordered r -tuple of multiplicatively independent integers of height at most H uniformly from the admissible tuples, admissibility excluding the Kummer-incompatible tuples defined below. In Programme IV, Conjectures 18, 19 sample the parameter c uniformly from the integers with $0 < |c| \leq H$ lying outside the postcritically finite and reducible loci, and Conjecture 17 uses the same law for its family-averaged clause. In Programme V the integer n is uniform on $\{1, \dots, N\}$. Conjecture 16 samples p uniformly from the primes $p \leq X$ that are unramified in the relevant field.

Prime motifs, the overlap poset and connected singular series

Throughout, p and q denote primes, $\pi(x)$ is the prime counting function, $\gamma = 0.5772156\dots$ is Euler’s constant, $\text{Li}(x) = \int_2^x dt/\log t$, and Λ is the von Mangoldt function. For an algebraic point, h denotes the absolute logarithmic Weil height and $\hat{h}_f$ the canonical height of a polynomial f . We write $\mathbb{E}_X$, Var_X , Cov_X for moments under the window-position law and $\mathbb{E}_Q$ for expectation under the modulus-class law.

A *motif* is a finite set $H = \{h_1 < \dots < h_k\} \subset \mathbb{Z}_{\geq 0}$ with $h_1 = 0$. It is *admissible* if for every prime p the reduction of H modulo p omits a residue class. With $\nu_H(p) = \#(H \bmod p)$ the singular series is

$$\mathfrak{S}(H) = \prod_p \frac{1 - \nu_H(p)/p}{(1 - 1/p)^k}.$$

Write $\Lambda_H(n) = \prod_{h \in H} \Lambda(n + h)$. For a smooth compactly supported w and $x \in \Omega_X$ the *centred weighted motif field* is

$$Y(H; x, L) = \sum_{n \in \mathbb{Z}} w\left(\frac{n-x}{L}\right) (\Lambda_H(n) - \mathfrak{S}(H)).$$

Under (W) this is a random variable, and $V = V(H; X, L) = \text{Var}_X Y(H; \cdot, L)$ is its *second connected mass*.

Let $H_1, \dots, H_m$ be motifs. A *configuration* of order r is a choice of indices $i_1, \dots, i_r$ together with shifts $t_1, \dots, t_r \in \mathbb{Z}$. Its *overlap type* is the set of coincidences

$$\tau = \{ \{ (u, h), (v, h') \} : u \neq v, h \in H_{i_u}, h' \in H_{i_v}, h + t_u = h' + t_v \},$$

recorded up to simultaneous translation. Overlap types are ordered by inclusion and form the *overlap poset* $\mathcal{O}(H_{i_1}, \dots, H_{i_r})$, whose minimum is the empty type (all translates disjoint) and whose grading $\text{codim } \tau$ is the number of independent linear relations among $t_1, \dots, t_r$ that τ imposes. The empty type is called the *disjoint stratum*.

For a finite family A of translated motifs let $\mathfrak{S}(A)$ be the singular series of the union pattern $\bigcup_{a \in A} a$, defined whenever that union is admissible. The *connected singular series* is the Möbius inversion of $\mathfrak{S}$ over the partition lattice $\Pi(A)$:

$$\mathfrak{S}^c(A) = \sum_{\pi \in \Pi(A)} \mu(\pi, \hat{1}) \prod_{B \in \pi} \mathfrak{S}(B), \quad \mu(\pi, \hat{1}) = (-1)^{|\pi|-1} (|\pi| - 1)!,$$

so that $\mathfrak{S}^c(A)$ vanishes whenever $\mathfrak{S}$ factorizes over a proper partition of A . This is the exact sense of “connected” used throughout Programme I, and it is what isolates genuine r -way arithmetic interaction from products of lower-order interactions.

For $r \geq 2$ and $j \geq 0$ the *incidence tensor* $A_r^{(j)}$ of the family is the symmetric r -th order form on $\mathbb{R}^m$ whose entry at $(i_1, \dots, i_r)$ is the total connected singular-series mass of the overlap types of codimension j carried by the r -fold configurations of $H_{i_1}, \dots, H_{i_r}$, weighted by the window integrals of Conjecture 1. The order of the tensor is the cumulant order r , whereas the superscript j indexes the strata by codimension and is not a tensor order, so the two indices never coincide in meaning. In particular at second order $A^{(j)} := A_2^{(j)}$ is a symmetric $m \times m$ matrix and $A^{(j)}c$ is a vector, for every j and not only for $j = 2$. Two translates carry a single shift parameter, so the overlap poset of a pair has exactly two levels, codimension 0, the disjoint stratum, in which the shift is free, and codimension 1, the coincidence strata, in which a coincidence pins the shift to one of finitely many values. Thus $j \in \{0, 1\}$ when $r = 2$. The coincidence strata are graded further by the coincidence multiplicity $e(\tau) \geq 1$ of Conjecture 1, and $A^{(1,e)}$ denotes the part of $A^{(1)}$ carried by the types with $e(\tau) = e$, so that the coincidence matrix of scale X is

$$A^{(1)}(X) = \sum_{e \geq 1} (\log X)^e A^{(1,e)}.$$

The disjoint stratum $j = 0$ carries no constant tensor. Its mass is the renormalized kernel matrix $K_{\text{HL}}(L)$ of Conjecture 3, which is a symmetric $m \times m$ matrix for each L but depends on the actual shift values and on L rather than on the overlap type alone, and it is placed last in the filtrations of Conjectures 2 and 5 because the annihilation conditions are imposed on the coincidence tensors first. “Last stratum” therefore refers to the order in which the strata are examined and not to the codimension grading, in which the disjoint stratum is the minimum.

Least primes and occupancy

For prime q and $a \in (\mathbb{Z}/q\mathbb{Z})^\times$ let $p(q, a)$ be the least prime congruent to a modulo q and set $T_q(a) = \text{Li}(p(q, a))/(q - 1)$. Under (M) this is a random variable. The *first-arrival transform* $\mathcal{F}$

has for domain the bounded measurable kernels $K: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ and for codomain the continuous functions on $\mathbb{R}_{\geq 0}$ vanishing to second order at the origin, and it is defined by the linearization

$$\mathcal{F}[K](t) = \frac{1}{2} \int_0^t \int_0^t K(|u-v|) du dv = \int_0^t (t-u)K(u) du.$$

This is the derivative at the constant unit intensity of the map that sends the connected two-point correlation of a point process on Li-time to minus the logarithm of its first-arrival survival function: a process of unit intensity and connected pair correlation K has $-\log \mathbb{P}[\text{no arrival in } [0, t]] = t - \mathcal{F}[K](t) + O(\|K\|_\infty^2)$. The *connected two-prime exclusion kernel* is the family of numbers $\mathfrak{S}(\{0, kq\}) - 1$ for $k \geq 1$, where $\mathfrak{S}(\{0, d\})$ is the singular series of the pair motif of shift d , these being the connected densities of two primes lying in the same class modulo q . Extending T_q to $a = 0$ by the recorded value at the zero class, the *additive spectrum* and *multiplicative spectrum* of the centred field $\tilde{T}_q(a) = T_q(a) - \mathbb{E}_a[T_q]$ are

$$S_q^{\text{add}}(\xi) = \frac{1}{q} \left| \sum_{a \bmod q} \tilde{T}_q(a) \exp(2\pi i a \xi / q) \right|^2, \quad S_q^{\text{mult}}(\chi) = \frac{1}{q-1} \left| \sum_{a \neq 0} \tilde{T}_q(a) \overline{\chi(a)} \right|^2,$$

for $\xi \in \mathbb{Z}/q\mathbb{Z}$ and χ a Dirichlet character modulo q . Both are deterministic for each q and become random variables under (M).

Class groups

For a negative fundamental discriminant D write $h(D)$ for the class number of $\mathbb{Q}(\sqrt{D})$ and $\text{Cl}(D)$ for the class group. The *two-primary genus data* of D is the pair consisting of the genus character group of D , equivalently the set of prime discriminant factors of D , together with the induced two-primary quotient of $\text{Cl}(D)$. For an odd prime ℓ , the *Cohen–Lenstra law* for $\text{Cl}(D)[\ell^\infty]$ assigns a finite abelian ℓ -group G the mass proportional to $1/|\text{Aut } G|$.

Finite logarithms, saturation and Kummer compatibility

For a prime p and $a \in \mathbb{Z}$ with $p \nmid a$ the *Fermat quotient* is $q_p(a) = ((a^{p-1} - 1)/p) \bmod p$. For a finitely generated $\Gamma \leq \mathbb{Q}^\times$ with p dividing no numerator or denominator of a generator, q_p extends to a homomorphism $q_p: \Gamma \rightarrow \mathbb{F}_p$.

Γ is *saturated* if $\Gamma = \{x \in \mathbb{Q}^\times : x^n \in \Gamma \text{ for some } n \geq 1\}$, and Γ^{sat} denotes the saturation. Saturation does not change $\Gamma \otimes \mathbb{Q}$, so $\text{rank } \Gamma = \text{rank } \Gamma^{\text{sat}} =: r$.

The *Kummer degree* of Γ at level m is

$$\Delta_\Gamma(m) = [\mathbb{Q}(\zeta_m, \Gamma^{1/m}) : \mathbb{Q}(\zeta_m)],$$

a divisor of m^r . Γ is *Kummer-generic at p* if $\Delta_\Gamma(p) = p^r$, and the *local Kummer factor* is

$$\kappa(\Gamma; p) = \frac{p^r}{\Delta_\Gamma(p)} \geq 1,$$

equal to 1 exactly at the Kummer-generic primes. A tuple of generators is *Kummer-compatible* if the group it generates is saturated and Kummer-generic at all but finitely many primes. Since $(\mathbb{Z}/p^2\mathbb{Z})^\times$ is cyclic of order $p(p-1)$, its subgroup of p -th powers is exactly $\{x : x^{p-1} = 1\}$, so

$$q_p(a) = 0 \iff a^{p-1} \equiv 1 \pmod{p^2} \iff a \bmod p^2 \text{ is a } p\text{-th power},$$

which is the identification of q_p with the connecting map $\Gamma \rightarrow (\mathbb{Z}/p^2\mathbb{Z})^\times / ((\mathbb{Z}/p^2\mathbb{Z})^\times)^p \cong \mathbb{Z}/p\mathbb{Z}$ used throughout Programme III.

Characters are taken on the ambient generator torus and not on Γ . Fix an ordered generating tuple $g = (g_1, \dots, g_s)$ of Γ , so that the Fermat-quotient vector

$$\mathbf{q}_p(g)/p = (q_p(g_1), \dots, q_p(g_s))/p \in (\mathbb{R}/\mathbb{Z})^s$$

lives on the ambient torus $(\mathbb{R}/\mathbb{Z})^s$. Let

$$R(g) = \left\{ m \in \mathbb{Z}^s : \prod_{i=1}^s g_i^{m_i} = 1 \right\}$$

be the *relation lattice* of the tuple, of rank $s - r$, let $R(g)^{\text{sat}} = (R(g) \otimes \mathbb{Q}) \cap \mathbb{Z}^s$ be its saturation in $\mathbb{Z}^s$, and let $\mathbb{T}_\Gamma \leq (\mathbb{R}/\mathbb{Z})^s$ be the identity component of the annihilator of $R(g)$, a subtorus of dimension exactly r . A *character* is an element $m \in \mathbb{Z}^s$, acting on the ambient torus by $\langle m, \mathbf{q}_p(g) \rangle = \sum_i m_i q_p(g_i)$, and its *height* is $\max_i |m_i|$. A character m is a *nonrelation character* of the tuple if $m \notin R(g)^{\text{sat}}$, equivalently if m restricts nontrivially to $\mathbb{T}_\Gamma$. The exclusion is exactly the right one, since for $m \in R(g)^{\text{sat}}$, say with $km \in R(g)$, one has $k\langle m, \mathbf{q}_p(g) \rangle \equiv 0 \pmod{p}$ and hence $\langle m, \mathbf{q}_p(g) \rangle \equiv 0 \pmod{p}$ at every $p \nmid k$, so such a character is constant on the values and carries no information, whereas for $m \in \text{Hom}(\Gamma, \mathbb{Z})$ the relations of Γ are killed automatically and no nontrivial exclusion is available at all.

Arboreal notation

Let $f \in \mathbb{Q}[x]$ have degree $d \geq 2$ and let $a \in \mathbb{Q}$ have infinite backward orbit. Write $K_n = \mathbb{Q}(f^{-n}(a))$ for the n -th preimage field, $G_n = \text{Gal}(K_n/\mathbb{Q})$, $D_n = \text{disc}(K_n)$, and T_n for the tree of preimages of a to level n , so that $G_n \hookrightarrow \text{Aut}(T_n) = [S_d]^n$, the n -fold iterated wreath product of the symmetric group S_d , of order $(d!)^{(d^n-1)/(d-1)}$. Only for $d = 2$ is this an iterated wreath product of cyclic groups. The *arboreal image* is $G_\infty = \varprojlim G_n \leq \text{Aut}(T_\infty)$, and it has *finite index* if $[\text{Aut}(T_\infty) : G_\infty] < \infty$. A *conjugacy cylinder* at level n is a conjugacy class of G_n , and a *cylinder event* is the corresponding set of factorization types of $f^j(x) - a$ modulo p for $j \leq n$. The *entropy-conductor functional* is

$$E(n, X) = \frac{|G_n|}{\pi(X)} + \frac{|G_n|^2 (\log D_n + [K_n : \mathbb{Q}] \log X)^2}{X},$$

the first term being the information-theoretic requirement that every state of G_n receive enough primes and the second the square-root Chebotarev conductor obstruction. A polynomial is *disintegrated* if it is not conjugate to a power map, a Chebyshev polynomial or a Lattès map, this being the class in which the Medvedev–Scanlon classification of invariant subvarieties applies.

For $f_c(x) = x^d + c$ put $v_n(c) = f_c^n(0)$. The *primitive part* $P_n(c)$ is the largest divisor of $|v_n(c)|$ composed only of primes dividing no $v_m(c)$ with $1 \leq m < n$.

Factor processes and Poisson–Dirichlet marks

PD(1) denotes the Poisson–Dirichlet law of parameter 1, the law of the decreasing rearrangement of the normalized jumps of a gamma subordinator, equivalently the limit law of the ranked ratios $\log p_i / \log n$ over the prime factors of a uniform integer $n \leq N$. Its largest coordinate has mean the Golomb–Dickman constant $\lambda = 0.6243299\dots$. Given a mark space S and a probability

kernel $\nu(\cdot \mid u)$ on S , a *marked PD(1) process* is the point process $\sum_i \delta_{(u_i, s_i)}$ on $(0, 1] \times S$ in which (u_i) is PD(1) and the marks s_i are conditionally independent given (u_i) with $s_i \sim \nu(\cdot \mid u_i)$. For irreducible f with splitting group G acting on the roots, $\text{fix}(C)$ is the number of roots fixed by any element of the conjugacy class C , and Burnside's lemma gives $\sum_C (|C|/|G|) \text{fix}(C) = 1$ for transitive G , so the mark intensity of Conjecture 21 needs no fitted normalization.

For a polynomial value $f(n)$ and a parameter y , the *y-smooth part* $S_y(f(n))$ is the largest y -smooth divisor and $R_y = f(n)/S_y(f(n))$ is the *residual cofactor*. The *small-prime Kubilius field* is the sigma-field generated by the valuations $v_p(f(n))$ for $p \leq y$. For irreducible f we write $\rho_f(p) = \#\{x \bmod p : p \mid f(x)\}$, $V_f(y) = \prod_{p \leq y} (1 - \rho_f(p)/p)$, $C_f = \prod_p (1 - \rho_f(p)/p)(1 - 1/p)^{-1}$ for the Bateman–Horn constant, and ω for the Buchstab function, the continuous solution of $(u\omega(u))' = \omega(u - 1)$ with $\omega(u) = 1/u$ on $1 < u \leq 2$. For several polynomials, $\rho(I, p) = \#\{x \bmod p : p \mid g_i(x) \text{ for all } i \in I\}$ is the *joint root-count tensor*.

Tagging conventions

Clauses marked **[elementary]** are proved by the argument indicated with them. Clauses marked **[conditional proposition]** are deductions from a hypothesis named in place. Clauses marked **[strong converse layer]** are exhaustiveness or completeness assertions. These are stated separately from the constructive clauses of the same conjecture and are separately falsifiable: a counterexample to a converse clause leaves the constructive clause of that conjecture standing, and the two therefore carry independent risk. Clauses marked **[stated programme]** record the derivation route for a constant or kernel that is canonical but not yet evaluated in closed form. All remaining clauses are conjectural. Questions raised in the QUESTION environment are open problems and are not counted among the twenty-five conjectures.

Evidence conventions

The evidence note attached to each conjecture reports a finite-range diagnostic and nothing more. Computations are consistency checks, never support for an asymptotic claim. Quoted standard errors and z -scores are descriptive, and where samples are correlated, overlapping or selected after inspection they are not calibrated tail probabilities. Where a range is too short to identify an asymptotic parameter this is said. All prime-pattern window statistics below were sampled at step $L/4$ within a window of length L , so consecutive samples overlap in three quarters of their support and the effective number of independent samples is about one quarter of the nominal count, with no error bars computed by the generating script. The generating script samples window positions on a uniform grid over $[X, 2X]$ rather than from the dyadic-log density of (W) , an admissible discretization of (W) because the two densities differ by a factor bounded between $1/(2 \log 2)$ and $1/\log 2$ on that interval, so no reported statistic is affected beyond that bounded reweighting. Reproducible bounds are primes to 3,000,000, prime moduli below 1000, Fermat quotients to 250,000, imaginary quadratic discriminants to 15,000, and the polynomial and dynamical samples recorded with each note.

2 Programme I. Connected prime-pattern fields

The programme builds a connected generating functional for prime motifs, an overlap filtration of its variance, a decomposition of its covariance into local and spectral halves, a conditioned extreme-

value theory, and a converse classification of low-variance combinations. All five statements live on the window-position probability space (W) of Section 1.

Conjecture 1 (Connected Hardy–Littlewood generating functional). *Fix admissible motifs $H_1, \dots, H_m$, a smooth compactly supported window w , and the window-position law (W). Write $Y_i = Y(H_i; \cdot, L)$ and let κ_r denote the joint cumulant of order r under $\mathbb{P}_X$. Then: (i) [overlap expansion] for every $r \geq 2$ and every choice of indices $i_1, \dots, i_r$ the cumulant $\kappa_r(Y_{i_1}, \dots, Y_{i_r})$ has an asymptotic expansion indexed by the overlap poset $\mathcal{O}(H_{i_1}, \dots, H_{i_r})$, in which each stratum carries an explicit power of $\log X$ and a renormalized kernel sum over its free shifts. For $\tau \in \mathcal{O}$ put $\text{fr}(\tau) = r - 1 - \text{codim } \tau$ for the number of free shifts of τ , and for $s \in \mathbb{Z}^{\text{fr}(\tau)}$ let $U(\tau; s)$ be the union pattern of the configuration of type τ with free shifts s , a set of distinct integers, and let $\mathfrak{S}^c(\tau; s) = \mathfrak{S}^c(U(\tau; s))$ be its connected singular series, obtained by Möbius inversion over the partition lattice as in Section 1. Let*

$$e(\tau) = \sum_{u=1}^r |H_{i_u}| - |U(\tau; s)|$$

be the coincidence multiplicity of τ , a quantity independent of s , so that a point of the union pattern carried by μ translates contributes $\mu - 1$ to $e(\tau)$. Then

$$\kappa_r(Y_{i_1}, \dots, Y_{i_r}) = \sum_{\tau \in \mathcal{O}} (\log X)^{e(\tau)} \mathcal{K}_\tau(L) + \text{lower strata},$$

$$\mathcal{K}_\tau(L) = \sum_{s \in \mathbb{Z}^{\text{fr}(\tau)}} w_L^{(\tau)}(s) \mathfrak{S}^c(\tau; s), \quad w_L^{(\tau)}(s) = L \int_{\mathbb{R}} w(t) \prod_{j=1}^{\text{fr}(\tau)} w\left(t + \frac{s_j}{L}\right) dt,$$

the window weight $w_L^{(\tau)}$ contributing one factor L for the summation over n and one for each free shift, so that $\mathcal{K}_\tau(L)$ sits at scale $L^{r - \text{codim } \tau}$ up to the logarithmic renormalization described in clause (v). The power of $\log X$ is the coincidence bookkeeping: a point of multiplicity μ contributes Λ^μ , and $\sum_{n \leq x} \Lambda(n)^\mu \sim x(\log x)^{\mu-1}$. In the smallest case $H = \{0\}$ and $r = 2$ the full-coincidence stratum has $e(\tau) = 1$ and $\text{codim } \tau = 1$ and contributes $L \log X$, matching $\sum_{n \leq x} \Lambda(n)^2 \sim x \log x$. (ii) [strong converse layer] no coefficient and no scale occurs outside this expansion: for every $s \geq 0$ the difference between κ_r and the expansion of (i) truncated at codimension s is $o(L^{r-s}(\log X)^{e_{\max}})$, where $e_{\max}$ is the largest coincidence multiplicity occurring at codimension at most s . Clause (ii) is separately falsifiable from (i), since a persistent normalized cumulant not assigned to any overlap stratum refutes (ii) while leaving the expansion of (i) intact. (iii) [hypothesis H_r] let $V = V(H_i; X, L)$ and let $\kappa_r^{\text{disj}} = \mathcal{K}_0(L)$ denote the contribution of the disjoint stratum, the renormalized kernel sum of (i) at $\text{codim } \tau = 0$. The hypothesis H_r , for $r \geq 3$, asserts that $\kappa_r^{\text{disj}} = o(V^{r/2})$ in every range in which $V \rightarrow \infty$. This is a hypothesis and not a consequence of (i) and (ii), which fix the shape of the expansion but not the size of its disjoint stratum. (iv) [conditional proposition, under H_r for every $r \geq 3$] $Y_i/\sqrt{V}$ converges in law under $\mathbb{P}_X$ to the standard Gaussian, and all higher joint cumulants of the normalized fields obey Wick pairing. Gaussianity is thus the conclusion drawn from clause (iii) and is not an independent input to the conjecture. (v) [renormalization of the free-shift strata] the free-shift strata, and in particular the disjoint stratum $\text{codim } \tau = 0$, are asserted only in the kernel-sum form displayed in (i), and not as a constant depending on τ alone times a window integral. The union pattern, and therefore $\mathfrak{S}^c(\tau; s)$, depends on the shift values s and not merely on the type, and the centred

shift sum diverges logarithmically, the two-motif case being

$$\sum_{0 < h \leq H} (\mathfrak{S}(\{0, h\}) - 1) = -\frac{1}{2} \log H + O(1),$$

in the form established by Montgomery and Soundararajan. Consequently $K_\tau(L)$ is smaller than $L^{r-\text{codim } \tau}$ by a logarithmic factor on the free-shift strata, and the kernel of the disjoint stratum at $r = 2$ is exactly the renormalized kernel K_{HL} of Conjecture 3.

Möbius inversion over set partitions is what isolates genuine r -way arithmetic interaction, since $\mathfrak{S}^c$ vanishes on any family whose singular series factorizes, while the overlap poset records the equal prime constraints that a partition cannot see. Together they play the role of connected diagrams in a cluster expansion, and the statement turns Hardy–Littlewood constants into the coefficients of a stochastic field theory for prime constellations rather than a list of separate predictions. If proved it would compute joint moments of motif counts, identify the exact obstruction to Gaussianity through clause (iii), and supply Conjectures 2 to 5 with a common parent instead of a family of analogies. The nearest antecedents are Gallagher’s averaging of singular series, the moment method of Montgomery and Soundararajan, the singular-series estimates of Pintz, and the odd-moment work of Kuperberg, each of which treats a projection of the object above rather than the full multitype functional.

Evidence. For four prime-pair fields at primes below 3×10^6 the standardized fourth connected cumulants lay between -0.31 and 0.41 and the selected mixed fourth cumulants between -0.13 and 0.26 as the window mean rose from 2.5 to 93 , but windows were sampled at step $L/4$ and therefore overlap in three quarters of their support, so about one quarter of the nominal samples are effective, no error bars accompany any of these numbers, and the range identifies no coefficient of the expansion.

Conjecture 2 (Overlap-filtration variance law). *Fix admissible motifs $H_1, \dots, H_m$, a window w and the law (W) . For $c \in \mathbb{R}^m$ put $Y_c = \sum_i c_i Y(H_i; \cdot, L)$. Then: (i) [variance filtration] $\text{Var}_X Y_c$ has an asymptotic expansion indexed by the overlap poset of the pairs (H_i, H_j) , which at second order has exactly the two levels recorded in Section 1. Grading the codimension-one strata by the coincidence multiplicity $e(\tau) \geq 1$ of Conjecture 1,*

$$\text{Var}_X Y_c = \sum_{e \geq 1} L(\log X)^e c^\top A^{(1,e)} c + c^\top K_{\text{HL}}(L) c + o(L),$$

in which $A^{(1,e)}$ is the symmetric $m \times m$ incidence matrix of Section 1 carrying the connected singular-series masses of the codimension-one overlap types of coincidence multiplicity e together with their window integrals, and $K_{\text{HL}}(L)$ is the renormalized disjoint kernel matrix of Conjecture 3, $K_{\text{HL}}(i, j) = \sum_h w_L(h) \mathfrak{S}^c(H_i, H_j + h)$. The disjoint stratum enters as a kernel sum over the actual shifts and not as a constant times L^2 , since the centred shift sum converges only after the logarithmic renormalization of Conjecture 1(v), and it carries no divergent mass. (ii) [annihilation, of depth one at second order] if $A^{(1,e)} c = 0$ for every $e \geq 1$ then the whole coincidence level vanishes from the expansion and the leading scale of $\text{Var}_X Y_c$ is $c^\top K_{\text{HL}}(L) c$, provided that quantity does not vanish. Because $j \in \{0, 1\}$ at $r = 2$, the annihilation chain has depth one: a second-order statistic admits exactly one nontrivial annihilation step, and any deeper cancellation is a condition on the higher-order tensors $A_r^{(j)}$ with $r \geq 3$ and is not visible in the variance alone. (iii) [strong converse layer] no variance scale occurs outside the coincidence matrices $A^{(1,e)}$ and the renormalized disjoint kernel, separately falsifiable from (i) and (ii) in the sense that an unassigned scale refutes (iii) alone.

Two occurrences of a motif may share $0, 1, 2, \dots$ prime constraints, and each sharing pattern contributes at its own power of the window length and its own power of $\log X$, so the overlap lattice is the arithmetic analogue of a diagram expansion and the variance is graded by it. The consequence is constructive: balanced observables can be designed to annihilate the low-codimension diagrams and thereby manufacture rigid prime statistics, which is the route to the arithmetic wavelets classified in Conjecture 5, and it would explain why apparently similar prime races fluctuate at different scales. Tuple correlations in the tradition of Hardy and Littlewood already distinguish coincident from disjoint shifts, and Gallagher's and Kuperberg's averaging arguments handle the extremes of the filtration, but a general variance grading indexed by the overlap poset is the content claimed here.

Evidence. The covariance matrix of four pair-pattern fields kept its three smaller eigenvalues at 0.29 to 0.44 of the largest with condition number near 3.4 across window lengths 400 to 15,000, which is consistent with a persistent mode hierarchy rather than a single common trend but identifies no stratum, and the overlapping-window caveat again applies, the step $L/4$ leaving about a quarter as many effective samples and no error bars.

Conjecture 3 (Local-to-spectral covariance decomposition for prime motifs). *Fix admissible motifs H_i , a smooth compactly supported window w , a fixed $\varepsilon > 0$ and the law (W) , and let $X^\varepsilon \leq L \leq X^{1-\varepsilon}$. Fix a modulus $Q_0 = Q_0(H_i, H_j)$ divisible by every prime $p \leq |H_i| + |H_j|$ and restrict n to a fixed class modulo Q_0 , so that each translated von Mangoldt factor $\Lambda(n+h)$ is expanded by the explicit formula in an arithmetic progression modulo Q_0 . Let $\mathcal{L}(H)$ be the resulting finite set of Dirichlet L -functions $L(s, \chi)$, χ modulo Q_0 , one for each character occurring in that expansion for the motif H , and write $\mathcal{Z}(H)$ for the multiset of nontrivial zeros of the members of $\mathcal{L}(H)$, each zero carried with the coefficient $\overline{\chi(a+h)}/\varphi(Q_0)$ with which it enters. Let*

$$M_{w,L,x}(s) = \int_0^\infty w\left(\frac{u-x}{L}\right) u^{s-1} du$$

be the window Mellin transform, so that the explicit formula gives

$$Y(H; x, L) = - \sum_{\rho \in \mathcal{Z}(H)} M_{w,L,x}(\rho) + \text{local terms.}$$

Let μ_{ij} be the joint zero-correlation measure of the pair, that is, the pair-correlation measure of the two zero multisets,

$$\mu_{ij} = \sum_{\rho \in \mathcal{Z}(H_i)} \sum_{\rho' \in \mathcal{Z}(H_j)} c_\rho \overline{c_{\rho'}} \delta_{(\rho, \rho')},$$

c_ρ denoting the coefficient carried by ρ . Then: (i) [decomposition] the motif covariance admits a unique asymptotic decomposition

$$\text{Cov}_X(Y(H_i; \cdot, L), Y(H_j; \cdot, L)) = D_{\text{ov}}(i, j) + K_{\text{HL}}(i, j) + K_{\text{spec}}(i, j) + o(V),$$

in which D_{ov} is the sum over the strata of positive codimension of the overlap poset of the pair, carrying the same logarithmic bookkeeping as Conjecture 1,

$$D_{\text{ov}}(i, j) = \sum_{e \geq 1} L(\log X)^e A_{ij}^{(1,e)},$$

K_{HL} is the renormalized connected kernel of the disjoint stratum, given by

$$K_{\text{HL}}(i, j) = \sum_h w_L(h) \mathfrak{S}^c(H_i, H_j + h), \quad w_L(h) = L \int_{\mathbb{R}} w(t) w(t + h/L) dt,$$

with w_L the window autocorrelation weight, and K_{spec} is the pullback of μ_{ij} through the window Mellin transforms,

$$K_{\text{spec}}(i, j) = \iint \text{Cov}_X(M_{w,L,\cdot}(\rho), M_{w,L,\cdot}(\rho')) d\mu_{ij}(\rho, \rho'),$$

the covariance being taken in the window position x under $\mathbb{P}_X$, so that K_{spec} is the window pullback of the joint zero-correlation measure and of nothing else. (ii) [strong converse layer] no fourth covariance source survives at leading order, that is, the residual after subtracting the three displayed terms is $o(V)$ for every admissible pair. This clause is separately falsifiable from (i), and a mixed term representable by neither connected local densities nor zero correlations refutes it alone. (iii) [crossover scale, derived] the crossover length $L_*(X)$ is defined by equality of the Hilbert–Schmidt norms of the two $m \times m$ matrix kernels,

$$\|K_{\text{HL}}(L)\|_{\text{HS}} = \|K_{\text{spec}}(L)\|_{\text{HS}}, \quad \|A\|_{\text{HS}}^2 = \sum_{i,j} |A_{ij}|^2,$$

this being the sense of “equal mass” used throughout, and $L_*(X)$ is therefore determined by (i) and not by a fitted exponent.

Local congruence correlations and global zero oscillations are not competing metaphors but two expansions of the same von Mangoldt field, and the coincident shifts must be separated first because they belong to neither description. A proof would join short-interval prime statistics, tuple singular series and zero correlations into one calculus, and would say when a random-sieve model suffices, when spectral information is unavoidable, and which residual statistic would signal genuinely new arithmetic structure. Explicit-formula methods and Hardy–Littlewood tuple expansions are the two established halves, developed for prime counts by Montgomery and Soundararajan and for the singular series by Pintz, and the claim distinctive here is that they are exactly the two pieces of one complete decomposition for arbitrary motifs.

Evidence. The twin-pair Fano factor fell from 0.83 to 0.71 as the window mean rose from 2.49 to 93.31 while the skewness moved toward zero, which is consistent with persistent sub-Poisson rigidity across scale but cannot separate the connected local kernel from a zero contribution at this range, and the windows again overlap at step $L/4$ so that roughly a quarter of the nominal samples are effective and no error bars exist.

Conjecture 4 (Spectrally conditioned extremal index for prime motifs). *Fix an admissible motif H and the window-position law (W) on $\Omega_X = [X, 2X]$ with window length L . Let $N_H(x)$ be the point process of occurrences of H in $[x, x + L]$, transformed by the exact cumulative Hardy–Littlewood hazard $t \mapsto \mathfrak{S}(H) \int_2^t (\log s)^{-|H|} ds$ so that it has unit intensity. Fix a cutoff wavelength L_{spec} with $L_{\text{spec}}/L \rightarrow \infty$, the symbol Λ being reserved for the von Mangoldt function, let $\Psi_{L_{\text{spec}}}(x)$ be the restriction to $[x, x + L]$ of the truncated explicit-formula sum over the zeros of the relevant L -functions with $|\Im \rho| \leq X/L_{\text{spec}}$, and let $\mathcal{F}_{L_{\text{spec}}} = \sigma(\Psi_{L_{\text{spec}}})$ be the sub-sigma-field of the Borel field of Ω_X that it generates.*

The admissible self-overlap graph $O(H)$ has for vertex set

$$V(H) = \{t \in \mathbb{Z} \setminus \{0\} : (H + t) \cap H \neq \emptyset \text{ and } H \cup (H + t) \text{ admissible}\},$$

a finite set contained in $(H - H) \setminus \{0\}$, and two vertices $t \neq t'$ are joined by an edge exactly when the three translates share a further coincidence and the triple union is admissible, that is, when

$$(H + t) \cap (H + t') \neq \emptyset, \quad \text{equivalently} \quad t - t' \in (H - H) \setminus \{0\},$$

and the union $H \cup (H + t) \cup (H + t')$ is admissible. For a set $S \subseteq V(H)$ write $\mathfrak{S}(S) = \mathfrak{S}(H \cup \bigcup_{t \in S} (H + t))$ for the Bateman–Horn density of the corresponding union pattern, so that $\mathfrak{S}(\emptyset) = \mathfrak{S}(H)$. Let $\mathcal{K}(H)$ be the set of connected components of $O(H)$ together with the empty component $\emptyset$, which records an isolated occurrence. The visited-component law is the probability measure on $\mathcal{K}(H)$ given by

$$\mathbb{P}[\mathcal{C}] = \frac{\mathfrak{S}(V(\mathcal{C}))}{\sum_{\mathcal{C}' \in \mathcal{K}(H)} \mathfrak{S}(V(\mathcal{C}'))},$$

the weight of a component being proportional to the Bateman–Horn density of the constellation it realizes, and the size of the visited component is $|V(\mathcal{C})| + 1$, the base occurrence included, so that the empty component has size one. Then: (i) the conditional law of the cluster starts of N_H given $\mathcal{F}_{L_{\text{spec}}}$ converges, as $X \rightarrow \infty$ and in probability under $\mathbb{P}_X$, to that of a homogeneous Poisson process. (ii) the within-cluster law is the visited-component law just defined, so the conditional extremal index is the reciprocal mean component size

$$\theta(H) = \left(\sum_{\mathcal{C} \in \mathcal{K}(H)} \mathbb{P}[\mathcal{C}] (|V(\mathcal{C})| + 1) \right)^{-1}$$

and involves no fitted parameter. (iii) after the conditioning is removed, that is, after averaging over $\mathcal{F}_{L_{\text{spec}}}$ under $\mathbb{P}_X$, the limit is the Cox cluster process directed by the exponential of the limiting low-mode field. (iv) [functoriality] if two motifs H and H' have isomorphic admissible self-overlap graphs, then $\theta(H) = \theta(H')$ once each process has been put on the unit-intensity clock by its own cumulative Hardy–Littlewood hazard, which is the local-density normalization meant here: the transform divides by $\mathfrak{S}(H)(\log t)^{-|H|}$, so the extremal index compares cluster structure and not the differing occurrence rates of the two motifs. Motifs with empty $O(H)$, that is with $V(H) = \emptyset$, admit only the empty component and have $\theta(H) = 1$. All four clauses are assertions about the probability space $(\Omega_X, \mathbb{P}_X)$. The spectral field is deterministic, what is random is the window position, and $\mathcal{F}_{L_{\text{spec}}}$ is the sigma-field the window position generates through the long-wavelength component.

Short overlaps create genuine clusters while zeros and other long modes modulate the local intensity, and conditioning separates the two mechanisms instead of absorbing both into a fitted location shift of a Gumbel law. The statement makes extreme prime-pattern gaps a probe of local arithmetic geometry and of the low-frequency prime field at once, predicting null cases with $\theta(H) = 1$, exact changes on adjoining a shift, and a principled source for apparent non-Gumbel mixtures. Generic extreme-value limits in the Gumbel class are classical and do not identify the arithmetic source of clustering, so the claimed content is the conditional cluster process together with the identification of its within-cluster law with the admissible self-overlap graph.

Evidence. For 17,937 twin-pair gaps in the upper part of the tested range the mean-scaled hazard had coefficient of variation 0.904 and Kolmogorov–Smirnov distance 0.060 from $\text{Exp}(1)$, and since the one per cent critical value at that sample size is about 0.012, the unconditioned hazard law is decisively non-exponential. The direction matters. A coefficient of variation of 0.904 is below one and is therefore underdispersion, whereas every mixture of exponentials, and in particular every Cox law of the form asserted in clause (iii), has coefficient of variation at least one. The measurement therefore argues against the bare exponential and against an unconditioned Cox reading alike, and is informative about rigidity at this range rather than about the mixture mechanism. No spectral conditioning is available here, so clauses (i) and (ii) are untested.

Conjecture 5 (Completeness of arithmetic wavelets). *Fix a finite family of admissible motifs $H_1, \dots, H_m$, the law (W) , and the incidence tensors $A_r^{(j)}$ of Section 1, together with the renormalized disjoint kernel K_{HL} , which is examined last. Write $\mathcal{A} = \bigcap_{e \geq 1} \ker A^{(1,e)}$ for the joint kernel of the second-order coincidence matrices. Then: (i) [constructive layer] if $c \in \mathbb{R}^m$ satisfies $A^{(1,e)}c = 0$ for every $e \geq 1$ and $c^\top K_{\text{HL}}(L)c \neq 0$, then $\text{Var}_X Y_c$ begins exactly at the renormalized disjoint kernel, with the scale and constant predicted by Conjectures 2 and 3, and if $c \notin \mathcal{A}$ then $\text{Var}_X Y_c$ begins at $L(\log X)^e c^\top A^{(1,e)}c$ for the largest e with $A^{(1,e)}c \neq 0$. (ii) [strong converse layer] conversely, every sequence of coefficient vectors whose variance is smaller than the generic coincidence scale $L(\log X)^{e_{\max}}$ by a fixed power of L or by a fixed power of $\log X$ converges in direction to $\mathcal{A}$. (iii) the space $\mathcal{A}$ is therefore the first canonical arithmetic wavelet space attached to the family, and the second-order filtration $\mathbb{R}^m \supseteq \mathcal{A}$ has depth one, in accordance with the two-level overlap poset of a pair. Its deeper terms are not visible in the variance and are cut out by the higher-order tensors, the space of level (r, s) being $\bigcap_{j \leq s} \ker A_r^{(j)}$ for $r \geq 3$, so that the arithmetic wavelet spaces of the family form a decreasing filtration indexed by the cumulant order as well as by the codimension. Clause (ii) is separately falsifiable from (i): a rigid coefficient sequence generated by a global spectral cancellation not represented in the finite overlap tensors refutes (ii) while leaving (i) and (iii) untouched.*

Cancellation of overlap diagrams is the arithmetic analogue of vanishing moments in wavelet theory, and the converse layer asserts that there is no other route to rigidity, so the finite overlap and spectral tensors exhaust the ways a balanced prime statistic can be made quiet. The practical consequence is that low-variance combinations can be constructed on purpose to expose deeper correlation strata, isolating zero terms, distinguishing competing tuple conjectures and giving high-sensitivity numerical tests. Balanced combinations of prime-counting functions are common in the literature, in the prime-race tradition surveyed by Martin and coauthors among others, but a converse classification of all unusually rigid finite motif statistics by overlap annihilation is the claimed content.

Evidence. The empirically smallest covariance mode of the four pair fields carried 0.58 to 0.59 of the mean coordinate variance at every window and its coefficient vector reached absolute alignment 0.987 between the two largest windows while remaining unstable at small windows, but that mode is selected by the same data that measure its variance, the windows overlap at step $L/4$ so that about a quarter of the nominal samples are effective, and no error bar accompanies the alignment.

Question 1. For four pair motifs, is the joint kernel $\mathcal{A} = \bigcap_{e \geq 1} \ker A^{(1,e)}$ of Conjecture 5 nonzero over $\mathbb{Q}$, and does the rational vector spanning it agree in direction with the empirically smallest covariance mode? This is an open question and is not counted among the twenty-five conjectures.

3 Programme II. Arithmetic occupancy and local-global decoupling

Least primes in residue classes form a field carrying both an additive exclusion geometry and a multiplicative spectrum, and odd class groups are tested against a growing set of local conditions. Conjectures 6 to 9 live on the modulus-class law (M), and Conjecture 10 on the uniform discriminant ensemble declared in its statement.

Conjecture 6 (First-order arithmetic occupancy kernel). *Work on the modulus-class law (M) and put $T_q(a) = \text{Li}(p(q, a))/(q - 1)$. Let $S_Q(t) = \mathbb{P}_Q[T_q(a) > t]$ be the survival function of the ensemble. Then, uniformly for t in compact subsets of $[0, \infty)$,*

$$S_Q(t) = e^{-t} \left[1 - \frac{A(t)}{\log Q} + o\left(\frac{1}{\log Q}\right) \right] \quad (Q \rightarrow \infty),$$

where A is the canonical function defined as follows. Expanding the empty-class probability $\mathbb{P}_Q[N_a(y) = 0]$, with $N_a(y) = \#\{p \leq y : p \equiv a \pmod{q}\}$, by inclusion and exclusion in connected prime correlations produces at second order the connected two-prime exclusion kernel $\mathfrak{S}(\{0, kq\}) - 1$, $k \geq 1$, of Section 1, and $A(t)$ is the coefficient of $1/\log Q$ obtained by applying the first-arrival transform $\mathcal{F}$ to that kernel and integrating to Li-time t . In particular A carries no fitted parameter and $A(0) = 0$. [stated programme] The closed evaluation of A from the numbers $\mathfrak{S}(\{0, kq\})$, the passage from the discrete kernel indexed by k to the bounded kernel on $\mathbb{R}_{\geq 0}$ on which the first-arrival transform $\mathcal{F}$ of Section 1 acts, which is the change of variable to Li-time, and the proof that all higher connected corrections are $O(1/\log^2 Q)$ after averaging q over the dyadic block, are the derivation route registered with this conjecture.

The independent collector gives the bare exponential, but nearby primes cannot occupy residue classes independently, since divisibility and prime-pair constraints impose a short-range exclusion process, and summing that exclusion produces the first arithmetic deformation of the survival law. A proof would supply the first quantitative stochastic theory of the whole least-prime field rather than an order of magnitude for the extreme class, and it feeds Conjectures 7 to 9 directly and generalizes to Chebotarev occupancy. Li, Pratt and Shakan formulate least primes as an arithmetic coupon collector and obtain the extremal lower bound in that language, and the content claimed here is the complete first-order survival deformation with its kernel derived from connected prime correlations.

Evidence. Across prime moduli from 101 to 999 the scaled mean speedup $(1 - \mathbb{E}_a[T_q]) \log q$ rose from 1.27 to 1.59 and the cover time sat 2.12 to 2.37 below the independent harmonic-mean value, so the sign and the persistence are inconsistent with an exchangeable collector at first order, but $\log q$ varies by less than a factor of 1.5 across the whole range, which cannot separate $1/\log Q$ from a $\log \log Q / \log Q$ correction.

Conjecture 7 (Dual additive-multiplicative spectral geometry of least primes). *Work on the modulus-class law (M), with $\tilde{T}_q$, S_q^{add} and S_q^{mult} as defined in Section 1. Then: (i) [additive half] for every fixed $\alpha \in (0, 1)$,*

$$\mathbb{E}_Q[S_q^{\text{add}}(\lfloor \alpha q \rfloor)] = \widehat{K}_{\text{add}}(\alpha) + o\left(\frac{1}{\log Q}\right),$$

where $\widehat{K}_{\text{add}}$ is the Fourier transform of $\mathcal{F}$ applied to the connected two-prime exclusion kernel, an explicit function carrying no fitted parameter. [stated programme] The closed evaluation of $\widehat{K}_{\text{add}}$ from the numbers $\mathfrak{S}(\{0, kq\})$ through the transform $\mathcal{F}$ of Section 1 is the derivation route registered here. (ii) [multiplicative half] characters of exact order k modulo q exist only when k divides $q - 1$, so the average is taken over the subensemble of (M) carried by the moduli $q \equiv 1 \pmod{k}$, renormalized to a probability measure on that subensemble, which is nonempty and of positive mass for every fixed k by Dirichlet's theorem. For every fixed order $k \geq 2$ the average of $S_q^{\text{mult}}(\chi)$ over the Dirichlet characters of exact order k modulo q equals $\lambda_k + o(1/\log Q)$ under that conditioned expectation, where λ_k is the explicit-formula contribution of the zeros of the

corresponding L -functions to the first-arrival functional. [stated programme] The closed evaluation of λ_k from those zeros through $\mathcal{F}$ is the derivation route registered here. (iii) both halves carry mass of the same order, so the field is diagonal in neither the additive nor the multiplicative basis, its covariance being the sum of two canonical operators of low complexity, one a convolution on the additive group of $\mathbb{Z}/q\mathbb{Z}$ and one diagonal in the Dirichlet characters. (iv) [strong converse layer] the two halves exhaust the field, in that $\mathbb{E}_Q[\|\tilde{T}_q\|_2^2]$ equals the total mass supplied by (i) and (ii) up to $o(1/\log Q)$, so no third invariant is required and the mixed additive-multiplicative correlations vanish after averaging q over the dyadic block. Clause (iv) is separately falsifiable from (i) to (iii): a persistent mixed component refutes (iv) alone and forces a third operator rather than a correction to the first two.

Prime differences see the additive group of $\mathbb{Z}/q\mathbb{Z}$ whereas the distribution among reduced classes sees its character group, and the first-arrival functional is nonlinear enough to retain both sources at first order without mixing them. A proof would give a genuine covariance theory for the entire least-prime landscape and would explain why purely independent, purely additive and purely prime-race models are each incomplete, and it supplies the environment conditioned on in Conjecture 8 and the background against which the spike of Conjecture 9 is defined. Prime races are expanded in multiplicative characters in the tradition surveyed by Martin and coauthors, and effective Chebotarev inputs of the kind refined by Thorner and Zaman govern the character side, while the exclusion between nearby primes is an additive phenomenon in the sense of Li, Pratt and Shakan, so the claim is that the least-prime field contains both geometries and that they exhaust its first-order covariance.

Evidence. The q -point transform on the full additive group, including the zero class, at which the recorded value is $p(q, 0) = q$ and therefore $T_q(0) = \text{Li}(q)/(q - 1)$, gave low-frequency to high-frequency power ratios of 1.13 to 1.42 across modulus bands, which rejects a flat spectrum and supports the additive half, while the multiplicative covariance was not estimated at all, so clause (i) is tested on one of its two operators and clause (iii) is untested.

Conjecture 8 (Conditioned Cox–Poisson law for the last residue classes). *Work on the modulus-class law (M) , so that the modulus q is uniform on the primes of $[Q, 2Q]$ and the centred arrival field $a \mapsto \tilde{T}_q(a)$ is a random element of $\mathbb{R}^{q-1}$. Say that a mode of $\tilde{T}_q$ is of terminal scale if its contribution to the variance of the field on the last-arrival window, that is to Var of $a \mapsto T_q(a) - \log(q - 1)$, stays bounded away from 0 and from ∞ as $Q \rightarrow \infty$. Conjecture 7 lists them explicitly: they are the additive frequencies ξ with $|\xi| \leq \xi_0$, where ξ_0 is the fixed cutoff beyond which $\widehat{K}_{\text{add}}(\xi/q)$ falls below a fixed positive threshold, together with the Dirichlet characters of exact order $k \leq k_0$, where k_0 is the fixed cutoff beyond which λ_k falls below the same threshold. The list is finite and does not depend on q . Let Z_q be the projection of $\tilde{T}_q$ onto those modes, a random vector on Ω_Q , and let $\mathcal{G}_Q = \sigma(Z_q)$ be the sub-sigma-field it generates. Let $c_6(q)$ be the deterministic centring supplied by Conjecture 6, namely the value that turns the survival law of that conjecture into a unit-rate exponential tail at the last-arrival level,*

$$c_6(q) = \log\left(1 - \frac{A(\log(q - 1))}{\log q}\right) = -\frac{A(\log(q - 1))}{\log q} + O\left(\frac{1}{\log^2 q}\right),$$

the extension of A from compact arguments to arguments of size $\log q$ being part of the stated programme registered with Conjecture 6. Let

$$\Xi_q = \sum_{a \in (\mathbb{Z}/q\mathbb{Z})^\times} \delta_{T_q(a) - \log(q-1) - c_6(q)}$$

be the point process of last-arrival excesses. Then: (i) conditionally on $\mathcal{G}_Q$, Ξ_q converges in law under $\mathbb{P}_Q$ to a homogeneous Poisson process of intensity $\theta_{\text{occ}} e^{-x} dx$. (ii) unconditionally, Ξ_q converges to the Cox process directed by $\theta_{\text{occ}} \exp(-x + W) dx$, where W is the limiting low-mode field of Conjecture 7. (iii) the constant θ_{occ} is the cluster functional of the short-range occupancy kernel of Conjecture 6 and is therefore derived rather than fitted. Every clause is a statement about the probability space $(\Omega_Q, \mathbb{P}_Q)$. The arrival field of a single modulus is deterministic, what is random is the modulus, and $\mathcal{G}_Q$ is the sigma-field generated by the finitely many mode coordinates of the sampled field.

Rare unfilled classes are approximately independent only after the slowly varying arithmetic environment is held fixed, since the low modes randomize the local intensity while the short exclusion kernel changes the extremal index, and the two effects are separated by exactly the conditioning declared above. The statement predicts the joint law of the last several classes, their spacings, multiplicities and character profiles, and it is a sharp diagnostic, because pure Poisson, Cox–Poisson and clustered Poisson models make different predictions for the probability of two exceedances. The classical coupon collector has a Poisson terminal process, and the point of departure is that the arithmetic field of Conjecture 7 has persistent low modes, so an unconditioned Poisson law is too rigid.

Evidence. The tested moduli are too small for the process-level claim, since the centred exceedance counts above 0, 1 and 2 were usually sparse or zero across the prime moduli below 1000, so clauses (i) and (ii) are untested and only the stable negative cover-time shift, which supports a nontrivial centring in clause (iii), was observed.

Conjecture 9 (Rank-one exceptional-zero deformation of occupancy). *Fix $\varepsilon_0 = 10^{-2}$ and let $\varepsilon \leq \varepsilon_0$ be fixed. Work on the modulus-class law (M) restricted to those q admitting a primitive real character χ_q whose L -function has a real zero β with $(1 - \beta) \log q \leq \varepsilon$, the limits being taken along any sequence of Q for which this subensemble is nonempty and carries positive mass. The statement is made for every fixed $\varepsilon \leq \varepsilon_0$, the value of ε_0 serving only to fix the meaning of “close to 1” and no clause depending on it otherwise. Under the generalized Riemann hypothesis the subensemble is empty for every Q , and all three clauses are then instanceless rather than false, which is why the conjecture is stated conditionally on the subensemble carrying positive mass. Then, after the ordinary occupancy kernel of Conjecture 6 is removed: (i) the centred first-arrival field has a deterministic component along χ_q whose amplitude is the first-arrival transform $\mathcal{F}$ applied to the exceptional explicit-formula term x^β/β , an explicit function of β and q with no fitted parameter. (ii) all modes orthogonal to χ_q obey Conjecture 8 to first order. (iii) [strong converse layer] no other first-order deformation survives, so the effect of an exceptional zero on the nonlinear first-arrival field is asymptotically of rank one. Clause (iii) is separately falsifiable from (i) and (ii), and higher harmonics of χ_q generated by the nonlinearity would refute it alone.*

The explicit formula inserts a single exceptional term proportional to $\chi_q(a)$, and linearizing the first-hit functional transfers that term to a rank-one perturbation of the field, in the manner of a spiked random matrix or a spiked random field. If true the statement makes least-prime occupancy a sensitive diagnostic for exceptional zeros and separates ordinary arithmetic dependence from zero-driven bias precisely, and if false the failure locates the nonlinearity that generates the missing harmonics. Effective Chebotarev and comparative prime number theory, in the forms developed by Thorner and Zaman and surveyed by Martin and coauthors, establish that real zeros create character biases in linear counts, and the new claim is that their effect on the nonlinear first-arrival field is of rank one.

Evidence. No support is available at the accessible scale, the Spearman correlation between the absolute quadratic-character projection and an inverse $L(1, \chi)$ proxy being 0.066 over the prime moduli below 1000, a null measurement rather than evidence against the conjecture, since no exceptional zero occurs in that range.

Conjecture 10 (Growing local-condition independence for odd class groups). *Fix an odd prime ℓ . Sample D uniformly from the negative fundamental discriminants with $|D| \leq Y$, and condition on the complete two-primary genus data of D in the sense of Section 1. Let $B = B(Y)$ and let $\Sigma_B(D) = ((D/p))_{p \leq B}$ be the vector of splitting signs. The single information measure used throughout is the number of imposed local bits $\pi(B)$, compared against the logarithmic information content $\log Y$ of the discriminant. Then: (i) whenever $\pi(B(Y)) = o(\log Y)$, the conditional law of $\text{Cl}(D)[\ell^\infty]$ given the genus data and given $\Sigma_B(D)$ converges in total variation, as $Y \rightarrow \infty$ and uniformly over the attainable sign vectors, to the Cohen–Lenstra law for the odd part conditioned on the genus data alone. (ii) the total-variation error is governed explicitly by $\pi(B)/\log Y$ and tends to zero whenever $\pi(B(Y)) = o(\log Y)$, which is the same range as in (i). No sharpness is claimed at the boundary $\pi(B) \asymp \log Y$, where the conjecture makes no assertion. (iii) the assertion is made for the full isomorphism type of $\text{Cl}(D)[\ell^\infty]$ and not only for the divisibility of $h(D)$ by ℓ , the latter being a coarse projection of the former.*

Genus theory is the known global coupling between local data and the class group, and once it is projected out, Frobenius data at bounded primes should consume only local entropy and should not bias the odd unramified extension group until the number of imposed conditions approaches the logarithmic information content of the discriminant. A proof would be a quantitative local-global decoupling principle joining Chebotarev data to Cohen–Lenstra statistics, would say how much local information a class group can absorb, and has direct analogues for Selmer groups and ray class groups. Wood establishes Cohen–Lenstra behaviour under fixed local conditions, Klagsbrun proves Davenport–Heilbronn theorems for quotients of class groups, Wang and Wood compute the moments that interpret the heuristics and Ellenberg surveys the field, and the content claimed here is uniformity over a growing set of Frobenius conditions after the genus obstruction is removed.

Evidence. For the 4560 negative fundamental discriminants with $|D| \leq 15,000$, conditioning on a single splitting sign at $p = 3, 5, 7, 11, 13$ moved the probability that 3 divides $h(D)$ by at most 0.012 and that 5 divides $h(D)$ by at most 0.026, but the test records divisibility of the class number and not the isomorphism type required by clause (iii), and the multi-cell variation is dominated by small samples.

4 Programme III. Finite logarithms, Kummer rank and algebraic tori

Finite logarithms live on relation tori, saturated rank and p -adic depth control the exceptional strata, and the same principle extends functorially from the multiplicative group to arbitrary algebraic tori. Every statement in this programme is asserted only for saturated groups and only after the Kummer degrees of Section 1 have been computed.

Conjecture 11 (Kummer–Haar law for Fermat-quotient homomorphisms). *Let $\Gamma \leq \mathbb{Q}^\times$ be finitely generated and saturated of rank r , fix an ordered generating tuple $g = (g_1, \dots, g_s)$ with relation lattice $R(g)$ and ambient torus $(\mathbb{R}/\mathbb{Z})^s$ as in Section 1, and let p range over the primes dividing*

no numerator or denominator of a generator. Then: (i) [elementary] $q_p(ab) \equiv q_p(a) + q_p(b) \pmod{p}$, so $q_p: \Gamma \rightarrow \mathbb{F}_p$ is a homomorphism and the ambient vector $\mathbf{q}_p(g)/p \in (\mathbb{R}/\mathbb{Z})^s$ lies in the subtorus $\mathbb{T}_\Gamma$ cut out by the relation lattice $R(g)$, a subtorus of dimension exactly r . (ii) as p varies through any fixed compatible Chebotarev class, defined by prescribed splitting behaviour in a fixed cyclotomic-Kummer field, $\mathbf{q}_p(g)/p$ equidistributes with respect to Haar measure on $\mathbb{T}_\Gamma$. (iii) [strong converse layer] the support is exactly $\mathbb{T}_\Gamma$ and no further constraint holds: for every nonrelation character $m \in \mathbb{Z}^s$ of the tuple, that is every m outside the saturation $R(g)^{\text{sat}}$, the mean of $\exp(2\pi i \langle m, \mathbf{q}_p(g) \rangle / p)$ over the class tends to zero. The exclusion of $R(g)^{\text{sat}}$ is necessary and sufficient, since on that sublattice the exponential is identically 1 at all but finitely many p , whereas every m outside it restricts nontrivially to $\mathbb{T}_\Gamma$. Clause (iii) is separately falsifiable from (ii), a persistent Fourier coefficient at a nonrelation frequency refuting it alone. (iv) [invariance, with the saturation caveat in force] the law depends on Γ only through Γ^{sat} , so two finitely generated groups with the same saturation give the same limit law and the controlling invariant is the saturated rank r and not the number of displayed generators. For a group that is not saturated, clause (iv) is false as stated with r read as the rank of the displayed generators, which is why saturation is part of the hypothesis rather than a proviso.

The Eisenstein–Lerch identity makes q_p a finite logarithm, Kummer theory identifies the exact relation lattice that its values must respect, and the conjecture asserts that no constraint beyond that lattice survives, so the limit is Haar on the quotient torus. A proof would supply a single structural law covering ordinary Fermat quotients, simultaneous near-Wieferich events, restrictions to arithmetic progressions and mod- p regulators, with rank replacing the number of bases as the controlling invariant. Shparlinski gives exponential-sum and value-set bounds for Fermat quotients and the horizontal statistics of Ostafe and Shparlinski describe the fixed- p picture, while the mod- p Leopoldt viewpoint of Boeckle, Guiraud, Kalyanswamy and Khare connects exceptional vanishing to regulators, and the full rank-torus law above is the claimed content.

Evidence. For the 22,041 primes $p \leq 250,000$ the exact relations $q_p(6) = q_p(2) + q_p(3)$ and $q_p(12) = 2q_p(2) + q_p(3)$ held to within 3×10^{-16} , which checks clause (i) and the implementation rather than the conjectural clauses, and the nontrivial one- and two-dimensional Fourier coefficients for bases 2, 3, 5 stayed below 0.011, of the square-root size $1/\sqrt{22041} \approx 0.0067$ expected for that sample, so clause (iii) is tested only at fixed frequencies.

Conjecture 12 (Square-root large sieve for finite logarithms). *For every saturated $\Gamma \leq \mathbb{Q}^\times$ of rank r and every ordered generating tuple $g = (g_1, \dots, g_s)$ of Γ there exists $\delta(\Gamma) > 0$ such that, for every nonrelation character $m \in \mathbb{Z}^s$ of the tuple in the sense of Section 1, that is every m outside the saturated relation lattice $R(g)^{\text{sat}}$, of height $\max_i |m_i|$ at most $X^{\delta(\Gamma)}$,*

$$\left| \sum_{X < p \leq 2X} e\left(\frac{\langle m, \mathbf{q}_p(g) \rangle}{p}\right) \right| \ll_{\Gamma, \varepsilon} \pi(X)^{1/2} X^\varepsilon,$$

uniformly after conditioning on fixed compatible cyclotomic and Kummer data, and the same bound holds for any fixed finite set of joint frequencies, the square-root scale being set by the number of prime samples and not by the ambient coordinate count. The growing height range is the essential content: the corresponding statement for a single fixed frequency is weak equidistribution, whereas the range $X^{\delta(\Gamma)}$ is what controls targets of shrinking measure.

Once the exact algebraic relations are quotiented out, finite logarithms should oscillate as genuinely independent phases, and the conjecture demands exactly the square-root cancellation

that independence predicts, neither more, which would be stronger than the model supports, nor less. A proof would be a large sieve theorem for p -adic logarithms, would make Conjecture 11 quantitative, and would yield the shrinking-target estimates that Conjectures 13 and 14 require and the toric analogues of Conjecture 15. The existing Fermat-quotient bounds of Shparlinski concern other averaging regimes and shorter frequency ranges, so the prime-varying, rank-sensitive and height-growing estimate above is the claimed strengthening.

Evidence. At frequencies 1, 2, 3, 5, 8, 13 the normalized Fourier magnitudes for bases 2, 3, 5 lay between 0.001 and 0.010 on about 22,000 primes, which is of the square-root size for that sample, but every frequency tested is fixed, so the computation bears on none of the growing-height content that distinguishes this statement from weak equidistribution.

Conjecture 13 (Saturated-rank threshold for simultaneous Wieferich primes). *Let $\Gamma \leq \mathbb{Q}^\times$ be finitely generated, let Γ^{sat} be its saturation and $r = \text{rank } \Gamma$, and let $\kappa(\Gamma; p) = p^r / \Delta_\Gamma(p) \geq 1$ be the local Kummer factor of Section 1. All clauses are asserted for $\Gamma = \Gamma^{\text{sat}}$ and only after the Kummer degrees $\Delta_\Gamma(p^k)$ have been computed. Put $W(\Gamma; X) = \#\{p \leq X : q_p \text{ vanishes on } \Gamma\}$. Then: (i) if $r = 1$ then $W(\Gamma; X) \sim \kappa(\Gamma) \log \log X$, with*

$$\kappa(\Gamma) = \lim_{X \rightarrow \infty} \frac{1}{\log \log X} \sum_{p \leq X} \frac{\kappa(\Gamma; p)}{p},$$

a compatible Kummer density determined by the degrees $\Delta_\Gamma(p)$ and equal to 1 when Γ is Kummer-generic at almost all p . (ii) if $r \geq 2$ then $\sum_p \kappa(\Gamma; p) p^{-r}$ converges and only finitely many such primes exist, the expected tail beyond Y being $\sum_{p > Y} \kappa(\Gamma; p) p^{-r}$. (iii) [invariance, with the saturation and compatibility caveats in force] adjoining to Γ a generator lying in $\Gamma \otimes \mathbb{Q}$ changes no term of (i) or (ii), because the saturation and hence r and every $\kappa(\Gamma; p)$ are unchanged, whereas adjoining a generator independent of Γ multiplies the local term at p by $1/p$ exactly at those p at which the enlarged group remains Kummer-generic, the factor being $\kappa(\Gamma'; p) / (p \kappa(\Gamma; p))$ at the finitely many entangled primes. Both halves of (iii) fail for a group that is not saturated, since the displayed rank is then not the exponent, and the second half fails for a Kummer-entangled extension, since the degree Δ is then not multiplied by p , so saturation and Kummer compatibility are hypotheses of the statement and not annotations to it.

Haar measure on an r -dimensional finite-logarithm torus assigns mass p^{-r} to the origin, so the divergence or convergence of the Borel–Cantelli sum is decided by the codimension alone, and Kummer splitting changes the local constant without touching that codimension. The statement unifies ordinary, simultaneous and relation-dependent Wieferich questions and gives an algebraic explanation of the threshold, and it is the direct arithmetic test of Conjecture 11. The one-base heuristic with probability $1/p$ is classical folklore, the mod- p Leopoldt framework of Boeckle, Guiraud, Kalyanswamy and Khare supplies the broader regulator setting, and Silverman derives non-Wieferich statements from the *abc* conjecture, with progressions by Graves and Murty, so the sharpening claimed here is that the saturated Kummer rank, and not the displayed number of bases, is the exact exponent.

Evidence. For symmetric near-zero boxes of half-width 0.02 the rank one, two and three counts were 922, 33 and 1 against Haar-volume predictions of about 882, 35 and 1.4, and exact vanishing occurred for individual bases but for no pair of bases below 250,000, which is compatible with clause (ii) at a range far too small to test a convergent tail.

Conjecture 14 (Rank-depth Poisson law for higher Wieferich strata). *Fix $r \geq 1$ and $\theta < 1$, and sample an ordered r -tuple of multiplicatively independent integers of height at most H uniformly*

from the admissible tuples, admissibility excluding the Kummer-incompatible tuples of Section 1. Let $\Gamma(H)$ be the saturation of the sampled tuple. Mark the pair (p, k) , for $p \leq H^\theta$ and $k \geq 1$, when every generator a satisfies $a^{p^{-1}} \equiv 1 \pmod{p^{k+1}}$, and put

$$k_{\max}(p) = \max\{k \geq 0 : (p, k) \text{ is marked}\},$$

with the convention that $(p, 0)$ is always marked, so that the marks are nested by construction and no Poisson law is available for the full set of marked pairs. Write

$$\kappa(r, k; p) = \lim_{H \rightarrow \infty} \mathbb{E}[p^{rk} / \Delta_{\Gamma(H)}(p^k)]$$

for the limiting expected local Kummer factor of the sampled tuple at depth k , computed from the Kummer degrees of Section 1 and equal to 1 when almost every sampled tuple is Kummer-generic at p . Existence of the limit, and hence independence of the intensity from the height parameter H , is part of the assertion. Then: (i) after the explicit Kummer local factors are inserted, the process of maximal-depth points $\{(p, k_{\max}(p)) : k_{\max}(p) \geq 1\}$ converges as $H \rightarrow \infty$ to a Poisson process on pairs (p, k) with intensity

$$\kappa(r, k; p) p^{-rk} - \kappa(r, k+1; p) p^{-r(k+1)},$$

equivalently, for each fixed $k \geq 1$ the depth- k slice $\{p : k_{\max}(p) \geq k\}$ converges to a Poisson process of intensity $\kappa(r, k; p) p^{-rk}$. The Poisson statement is made for the maximal-depth process and for the individual depth slices, and not for the nested family of marked pairs. (ii) conditionally on a depth- k mark at p , the depth- $(k+1)$ mark is a Haar lift imposing r further linear conditions, so the depths form a codimension- r tower rather than an independent sequence, which is the deterministic nesting recorded above. (iii) for a fixed Γ the same intensities give the Borel–Cantelli predictions of Conjecture 13. Clause (iii) is a conjecture and not a consequence of (i), since a single fixed Γ is not a sample from the height ensemble and carries no probability space of its own.

The p -adic logarithm of r independent generators is a vector in a rank- r lattice and each further digit imposes another codimension- r condition, while the height ensemble supplies the probability space that a fixed group cannot. The statement predicts all higher-depth exceptional strata, the dependence between depths, and the law of the finite exceptional set when $rk > 1$, and it connects Wieferich primes to the theory of random p -adic Smith normal forms. Higher Wieferich congruences are usually treated one base and one depth at a time, in the regulator language of Boeckle, Guiraud, Kalyanswamy and Khare or the *abc*-conditional language of Silverman and of Graves and Murty, and the joint rank-by-depth point process with a declared family ensemble is what is claimed here.

Evidence. No base-2 prime of depth at least two occurs below 250,000 while the depth-one primes are 1093 and 3511, and since $\sum_{p \geq 3} p^{-2} \approx 0.202$ the stated intensity predicts about 0.2 such primes in that range, so the observation is compatible with clause (i) and tests nothing about the family ensemble on which the conjecture is built.

Conjecture 15 (Toric finite logarithm and codimension principle). *Let $T/\mathbb{Q}$ be an algebraic torus with connected Néron model over $\mathbb{Z}[1/N]$, let $P \in T(\mathbb{Q})$, and let S be the identity component of the Zariski closure of the subgroup generated by P , of dimension r . Write $\mathcal{T}$ for the connected Néron model. A prime p is good if $p \nmid N$, if p is unramified in the splitting field of T , and if P is integral at p , that is if $P \in T(\mathbb{Z}[1/N])$ so that p divides no denominator of P and P extends*

to a section of $\mathcal{T}$ over $\mathbb{Z}_{(p)}$. For a good prime p let N_p be the exponent of $T(\mathbb{F}_p)$. Then $[N_p]P$ reduces to the identity modulo p , so its image in $\mathcal{T}(\mathbb{Z}/p^2\mathbb{Z})$ lies in the kernel of reduction, and the first finite logarithm $\ell_p(P)$ is defined as its image under the canonical isomorphism

$$\ker(\mathcal{T}(\mathbb{Z}/p^2\mathbb{Z}) \longrightarrow \mathcal{T}(\mathbb{F}_p)) \xrightarrow{\sim} \mathrm{Lie}(\mathcal{T})(\mathbb{F}_p) = \mathrm{Lie}(T)(\mathbb{F}_p),$$

the isomorphism of the first congruence neighbourhood of the identity with the Lie algebra, which for $T = \mathbb{G}_m$ sends $1 + pu$ to u and recovers the Fermat quotient. Then: (i) [elementary] $\ell_p(P) \in \mathrm{Lie}(S)(\mathbb{F}_p)$, so the algebraic relations satisfied by P annihilate the corresponding tangent coordinates identically. (ii) conditional on any compatible Frobenius class in the splitting field of T , the normalized vector $\ell_p(P)/p$, read in the reduction of a fixed $\mathbb{Z}$ -basis of $\mathrm{Lie}(S)$, equidistributes with respect to Haar measure on $(\mathbb{R}/\mathbb{Z})^r$. (iii) [functoriality] for a morphism of tori $\varphi: T \rightarrow T'$ one has $\ell_p(\varphi(P)) = d\varphi(\ell_p(P))$ at every good prime for both, so the construction and the law are functorial and the codimension invariant is geometric rather than a coordinate count. (iv) vanishing of $\ell_p(P)$ to p -adic depth k has local density p^{-rk} times the explicit splitting factor determined by the decomposition of p in the splitting field of T , the density being that of the target inside $\mathrm{Lie}(S)(\mathbb{Z}/p^k\mathbb{Z})$ under Haar measure. This is a statement about the measure of the target and not a probability over a declared ensemble of primes, no such ensemble being attached to a fixed P , and the divergent-summable boundary of the associated Borel–Cantelli sum $\sum_p p^{-rk}$ therefore depends only on $\dim S$ and k . (v) the case $T = \mathbb{G}_m$, and more generally the case of powers of $\mathbb{G}_m$, recovers Conjectures 11 and 13. The construction depends on the integral model, and the statement is asserted for the connected Néron model at good primes, where the component group is trivial.

The ordinary Fermat quotient is the case of the multiplicative group, and replacing coordinate counting by the tangent space of the algebraic subgroup generated by the point turns the exceptional exponent into a geometric invariant that transforms correctly under morphisms. A proof would extend Wieferich and mod- p regulator heuristics to norm-one tori, to units in number fields and to higher-dimensional algebraic groups, and it predicts which relations survive base change and which exceptional prime problems are genuinely of higher rank. The finite-logarithm picture for the multiplicative group is due in its analytic form to Shparlinski and in its regulator form to Boeckle, Guiraud, Kalyanswamy and Khare, and the extension to arbitrary tori with dimension and functoriality controlling the exponent is the broader framework claimed here.

Evidence. For the norm-one torus of $\mathbb{Q}(\sqrt{2})$ with $P = 3 + 2\sqrt{2}$ and 22,041 primes, the scalar tangent component vanished identically as the norm relation of clause (i) requires, the fixed Fourier magnitudes of the one-dimensional tangent coordinate lay between 0.0034 and 0.0099, the split and inert prime means were 0.4975 and 0.5000, and exact toric vanishing occurred at $p = 13$ and $p = 31$, all of which are fixed-frequency consistency checks on clause (ii) for a single point on a single torus.

Question 2. For a norm-one torus of a real quadratic field, is the local splitting factor of Conjecture 15(iv) at inert primes, where $|T(\mathbb{F}_p)| = p + 1$, equal to the split-prime factor, or does the difference of group orders shift it? This is an open question and is not counted among the twenty-five conjectures.

5 Programme IV. Arboreal resolution and primitive dynamical factors

An entropy-conductor boundary controls how much of an infinite arboreal representation is visible below a given height, and the primitive factors of orbit terms and the gcds between orbits are governed by arboreal and geometric structure. The family-averaged clauses of this programme use the parameter law (P) of Section 1.

Conjecture 16 (Joint arboreal Chebotarev law at growing depth). *Let $f \in \mathbb{Q}[x]$ be a non-postcritically-finite polynomial of degree $d \geq 2$ and let $a \in \mathbb{Q}$ have infinite backward orbit and arboreal image G_∞ of finite index in $\text{Aut}(T_\infty)$. Let $n = n(X)$ satisfy $E(n, X) \rightarrow 0$ in the notation of Conjecture 17, and sample p uniformly from the primes $p \leq X$ unramified in K_n . Then: (i) the complete compatible vector of factorization types of $f^j(x) - a$ modulo p for every $j \leq n$ converges in total variation to the Haar–Chebotarev distribution on conjugacy-cylinder data of G_n . (ii) every statistic obtained by projecting the tree, in particular root counts at each level, cycle lengths, first-preimage levels and cross-level coincidence events, converges jointly and uniformly in the same regime, so the convergence is at the level of the whole visible tree and not one quotient at a time.*

Chebotarev supplies one fixed quotient at a time, and the content here is that the entire visible finite tree stays equidistributed until the combined sample and conductor complexity reaches the critical value identified in Conjecture 17. A proof would turn an infinite profinite representation into a quantitatively observable arithmetic process, transferring group martingales and cycle statistics to growing finite-field depth, and it is the foundation on which Conjectures 18 to 20 rest. Arboreal representations control fixed-level factorization and prime-divisor densities in the work surveyed by Jones and in the results of Doyle, Healey, Hindes and Jones, with fixed-point proportions computed by Juul, Kurlberg, Madhu and Tucker and by Fariña-Asategui and Radi, and the joint law for all levels simultaneously in the maximal resolvable range is the claimed content.

Evidence. For quadratic maps $f_c(x) = x^2 + c$ with c drawn from a list of eleven parameters and primes $7 \leq p \leq 499$, the root-count distributions at levels 2, 3 and 4 lay within total variation 0.042 to 0.048 of the full binary wreath-product model, but the census covers ten effective maps rather than eleven, because the good-prime filter used, $p \not\equiv 0 \pmod{c}$, removes the parameter $c = 1$ entirely and otherwise deletes only the single prime $p = 7$ at $c = \pm 7$, and the test is fixed-level and therefore subcritical.

Conjecture 17 (Sharp entropy-conductor barrier for growing-level Chebotarev). *Let K_n , G_n , D_n and $E(n, X)$ be as in Section 1, and let $\text{TV}(n, X)$ denote the total variation distance between the empirical frequency vector of conjugacy cylinders up to level n , computed over the primes $p \leq X$ unramified in K_n , and the Haar–Chebotarev vector. Write*

$$D_n(f, X) = \max_C \left| \frac{\#\{p \leq X \text{ unramified in } K_n : \text{Frob}_p \in C\}}{\#\{p \leq X \text{ unramified in } K_n\}} - \frac{|C|}{|G_n|} \right|$$

for the maximal cylinder discrepancy at level n , the maximum being over the conjugacy cylinders C of G_n . Call $(f_c)_{|c| \leq H}$ a one-parameter family of finite-index type if every f_c has degree $d \geq 2$, if the basepoint has infinite backward orbit for every c outside a set of density zero in the parameter law (P), and if for every such c the arboreal image $G_\infty(f_c)$ has finite index in $\text{Aut}(T_\infty)$ with the index bounded uniformly in c . Then: (i) [upper half, conditional proposition

under the generalized Riemann hypothesis for the Dedekind zeta functions of the fields K_n] one has $\text{TV}(n, X) \ll E(n, X)^{1/2}(\log X)^{O(1)}$, the first term of E supplying the multinomial sampling error over $|G_n|$ states and the second the effective Chebotarev error summed over states, so $E(n, X)(\log X)^{O(1)} \rightarrow 0$ implies $\text{TV}(n, X) \rightarrow 0$. The logarithmic factor is not decorative. The effective Chebotarev bound under a Riemann hypothesis carries powers of $\log D_n$ and of $\log X$, and the second term of E records the square-root conductor obstruction without them, so clause (i) is asserted only up to factors of that shape. The sharpness pairing with clause (ii) is unaffected, since the two clauses are separated by a power of E and not by a logarithm. (ii) [lower half, family-averaged] let $(f_c)_{|c| \leq H}$ be a one-parameter family of finite-index type in the sense above, with c sampled by the parameter law (P) , and suppose $E(n, X) \rightarrow \infty$ along a sequence of pairs (n, X) . Then the family average $\mathbb{E}_c[D_n(f_c, X)]$ of the maximal cylinder discrepancy at level n stays bounded below by a positive absolute constant along that sequence. The clause is stated as a family average because a single fixed f produces a deterministic sequence of discrepancies for which no probability space is declared, and an “infinitely often” assertion for one f is therefore not the form asserted here. (iii) [critical window, conjectural profile] in the window $E(n, X) \asymp 1$ the centred and normalized empirical cylinder-frequency field has a Gaussian-multinomial limit whose covariance is the multinomial covariance projected by the tree quotient relations. This clause is the most speculative of the three and no finite computation below bears on it. The sharpness assertion of this conjecture is the pairing of (i) with (ii): the analytic input makes the subcritical side a deduction, and the conjectural content is that the same functional governs the supercritical side on average over a family.

The first term of $E(n, X)$ is the information-theoretic requirement that every state of G_n receive enough primes, the second is the square-root Chebotarev and conductor obstruction, and the claim is that their maximum rather than the tree level alone is the canonical resolution parameter. A sharp barrier would say exactly how much of an infinite arboreal representation can be observed below a given height, would organize growing-level questions throughout arithmetic dynamics, and would separate group growth from ramification growth as sources of irresolvability. The effective Chebotarev bounds that make clause (i) a deduction are those of Thorner and Zaman in the tradition of Lagarias and Odlyzko, while arboreal group calculations in the surveys of Jones and in Doyle, Healey, Hindes and Jones fix the growth of $|G_n|$, and the identification of the maximal resolvable depth is the claimed content.

Evidence. Only levels 2 to 4 were computationally accessible, over ten effective quadratic maps rather than the eleven parameters listed, since the good-prime filter removes $c = 1$ entirely, and their root-count distributions lay within total variation 0.042 to 0.048 of the wreath-product law, which is subcritical consistency alone and bears on neither the sharpness pairing of clauses (i) and (ii) nor the critical profile of clause (iii).

Conjecture 18 (Poisson–Dirichlet law for primitive factors in dynamical families). *Fix $d \geq 2$ and sample c by the parameter law (P) , uniformly from the integers with $0 < |c| \leq H$ outside the postcritically finite and reducible loci. Let $n = n(H)$ grow so slowly that $d^n = o(\log H)$, and let $P_n(c)$ be the primitive part of $v_n(c) = f_c^n(0)$ in the sense of Section 1, that is, the largest divisor composed of primes dividing no earlier orbit term. Then: (i) the ranked ratios $\log \ell_i / \log P_n(c)$, over the primes ℓ_i dividing $P_n(c)$ counted with valuation, converge in distribution as $H \rightarrow \infty$ to $\text{PD}(1)$. (ii) the limit is jointly marked by the level of first appearance of each prime and by its Frobenius data in the arboreal tower of Conjecture 16, the marks being conditionally independent given the masses. The growth condition $d^n = o(\log H)$ and the family sampling are part of the*

statement, since a fixed parameter would give a deterministic sequence of factorizations with no probability space attached.

Once the inherited factors forced by rigid divisibility are deleted, the primitive part should behave as a well-distributed large integer with no further macroscopic algebraic decomposition, and PD(1) is the canonical scale-invariant factor process for such an integer. A proof would connect arithmetic dynamics to modern factor-process theory and would quantify far more than the existence of a primitive divisor, predicting largest factors, smoothness and the number of macroscopic new primes. Primitive divisors and average Zsigmondy sets are established themes, in Zsigmondy's classical theorem and in the work of Ingram and Silverman and of Laishram, Rout and Yadav, and large prime factors of well-distributed sequences carry Poisson–Dirichlet behaviour by Bharadwaj and Rodgers, so the claimed content is the PD(1) factorization of the primitive part in a dynamical family with arboreal marks.

Evidence. For 158 parameters at level 4 the mean largest logarithmic share was 0.684 with median 0.658 against the Golomb–Dickman mean 0.6243 for PD(1), a gap of several standard errors of the mean if the parameters are treated as independent samples, so the finite-level structure is visible rather than absent, and level 4 with $d^n = 16$ against $\log H \approx 4.4$ is far outside the stated growth regime.

Conjecture 19 (Euler law for squarefull primitive orbit factors). *In the family and growth regime of Conjecture 18, let $\delta_p(n)$ be the density of residues $c \bmod p^2$ for which $p^2 \mid v_n(c)$ while p divides no earlier orbit term, a quantity computed by counting parameters modulo p^2 and therefore derived rather than fitted. Let the critical-orbit collision locus at level n be the set of parameters*

$$\mathcal{R}_n = \{c : v_i(c) \equiv v_j(c) \pmod{p^2} \text{ for some } 1 \leq i < j \leq n \text{ and some } p \mid P_n(c)\},$$

on which two distinct forward images of the critical point 0 agree to the second power of a primitive prime and the codimension-two congruence defining a repeated primitive factor degenerates. Then: (i) *the sequence $\delta_p(n)$ stabilizes, in the sense that there is $n_0(p)$ with $\delta_p(n) = \delta_p$ for every $n \geq n_0(p)$, and with that stable value the probability that $P_n(c)$ is squarefree converges to $\prod_p(1 - \delta_p)$.* (ii) *after conditioning on the complement of $\mathcal{R}_n$, the valuations $v_p(P_n(c))$ at distinct primes are asymptotically independent.* (iii) *the logarithm of the squarefull part of $P_n(c)$ is tight as $H \rightarrow \infty$.*

A repeated primitive factor is a codimension-two congruence condition on the parameter unless ramification in the critical orbit lowers the codimension, so the exceptional ramified loci must be separated explicitly before the Euler product can be asserted. The statement supplies the p -adic input that makes Conjecture 18 sound, since a systematic square factor would distort the mass normalization, and it refines dynamical Zsigmondy theory from an existence statement to a multiplicity statistic. Squarefree-value heuristics are classical, and the primitive and inherited decomposition together with the critical-orbit correction is what makes the law specifically dynamical, in the setting of Ingram and Silverman and of Laishram, Rout and Yadav.

Evidence. At level 4, 155 of the 158 tested primitive parts were squarefree, a fraction 0.981, and the three repetitions are recorded rather than discarded, but 158 samples at a single level cannot distinguish the conjectured Euler product from nearby constants and give no information about clauses (ii) and (iii).

Conjecture 20 (Geometric-arboreal criterion for large dynamical gcds). *Let $f, g \in \overline{\mathbb{Q}}[x]$ be disintegrated of degree at least two and let a, b have positive canonical height for f and g*

respectively. For $\alpha, \beta \in \overline{\mathbb{Q}}^\times$ lying in a number field K , the logarithmic gcd is Silverman's generalized logarithmic greatest common divisor

$$\log \gcd(\alpha, \beta) = \sum_{v \in M_K^0} n_v \min\{v^+(\alpha), v^+(\beta)\}, \quad n_v = \frac{[K_v : \mathbb{Q}_v]}{[K : \mathbb{Q}]},$$

the sum being over the finite places of K with $v^+(x) = \max\{0, -\log |x|_v\}$, a quantity independent of the field K containing α and β and equal to $\log \gcd(a, b)$ for positive integers. Consider

$$\limsup_{\min(n, m) \rightarrow \infty} \frac{\log \gcd(f^n(a), g^m(b))}{\max(h(f^n(a)), h(g^m(b)))} > 0,$$

the limit superior being taken over all pairs (n, m) of positive integers with $\min(n, m) \rightarrow \infty$, so that both orbits advance. Then: (i) [constructive direction] if, after replacing f and g by iterates, the orbit pair lies on a nontrivial periodic algebraic correspondence for (f, g) , equivalently if f and g have a common semiconjugate dynamical quotient carrying the branches of a and b , then the displayed limsup is positive. (ii) [strong converse layer] conversely, positivity of the limsup forces the existence of such a correspondence, so no accidental congruence source of macroscopic gcds exists. Clause (ii) is separately falsifiable from (i), and a thin infinite family of large gcds produced by a correspondence visible only after base change, or by a nonperiodic special subvariety, refutes it alone. (iii) [functoriality] such a correspondence induces a common quotient of the distinguished arboreal representations of f and g . (iv) in the absence of a correspondence the normalized gcd is $o(1)$ outside a set of pairs (n, m) of density zero, and the deterministic second-moment bound

$$\frac{1}{N^2} \sum_{n, m \leq N} \left(\frac{\log \gcd(f^n(a), g^m(b))}{\max(h(f^n(a)), h(g^m(b)))} \right)^2 \ll \frac{1}{\log N}$$

holds as $N \rightarrow \infty$. The bound is deterministic and is asserted over the pairs (n, m) directly. The pairs carry no sampling law, so no covariance, connected or otherwise, is attributed to the sequence of common primes.

A common prime is a Frobenius event in the fibre product of the two preimage towers, but only the common quotients that arise from geometry persist strongly enough to create macroscopic gcds, which is why a criterion phrased through finite quotients alone is too broad. An equivalence would connect dynamical unlikely intersections, gcd bounds and profinite Galois representations in one statement, and either outcome identifies the responsible invariant, geometry rather than accidental congruence. Orbit gcds, invariant curves and arboreal entanglement are studied separately in the work of Ingram and Silverman, of Asgarli and coauthors on collisions of orbits over function fields, and in the classification of invariant varieties by Medvedev and Scanlon, which is what makes the disintegrated hypothesis the right one. The gcd bounds for values of algebraic functions are those of Bugeaud, Corvaja and Zannier, and the dynamical gcd conjectures of Silverman are the immediate predecessors of the constructive direction, so the claimed content is that the geometric correspondence is the complete source.

Evidence. For small quadratic maps at levels 2 to 6 the independent-parameter pairs had mean maximum normalized logarithmic gcd 0.035 with maximum 0.069 whereas deliberately related same-orbit pairs averaged 0.097 and reached 0.246, which illustrates the predicted separation over nine parameters and five levels and tests neither direction of the equivalence.

6 Programme V. Frobenius-marked polynomial factorization

Macroscopic factors of polynomial values carry Galois marks, the small and large scales decouple once the correct normalization is used, reducible polynomials glue their colour processes through an explicit local factor, and smoothness and primality are joined by a sieve flow. Every statement uses the uniform law on $n \leq N$ declared in Section 1.

Conjecture 21 (Frobenius-marked Poisson–Dirichlet factor process). *Let $f \in \mathbb{Z}[x]$ be irreducible with splitting field L and splitting group $G = \text{Gal}(L/\mathbb{Q})$ acting transitively on the roots. Sample n uniformly from $\{1, \dots, N\}$, mark each prime factor p of $f(n)$ by its Frobenius conjugacy class C in G , and give it mass $u = \log p / \log |f(n)|$. Then: (i) the macroscopic marked process converges as $N \rightarrow \infty$ to a marked PD(1) process whose mark intensity is proportional to $(|C|/|G|) \text{fix}(C) du/u$, the normalization being exactly one by Burnside’s lemma, so no constant is fitted. (ii) for every fixed number of macroscopic factors, their Frobenius marks are asymptotically independent conditional on their masses, once the finite local congruence state of n is removed. (iii) [functoriality] the law is functorial under quotienting the splitting group, the marked process for a quotient G/H being the pushforward of the marked process for G .*

A prime divides a value of f with frequency equal to its number of roots modulo p , so Chebotarev size-biases the conjugacy classes by fixed-point counts while scale invariance of the factorization produces the Poisson–Dirichlet masses, and the two mechanisms compose without a free parameter. A proof would unify large-factor statistics with Galois theory, predicting not only how large the macroscopic factors are but how they split in the defining field, and it extends naturally to covers, norm forms and arithmetic monodromy families. Large prime factors of sufficiently well-distributed sequences have Poisson–Dirichlet behaviour by Bharadwaj and Rodgers, with the prime-value side developed by Kravitz, Woo and Xu, and the simultaneous Galois mark on every macroscopic factor is the claimed content.

Evidence. For $f(x) = x^3 - 2$ and $n \leq 1200$ the largest prime factor had one root modulo p in 50.58 per cent of samples and three roots in 49.42 per cent against the size-biased prediction of one half and one half for S_3 , a deviation of 0.58 percentage points against a descriptive standard error of about 1.4 percentage points at that sample size, so the comparison is a consistency check at a single degree of freedom and bears on neither clause (ii) nor clause (iii).

Conjecture 22 (Small-large factor independence after logarithmic mass removal). *Let $f \in \mathbb{Z}[x]$ be irreducible, sample n uniformly from $\{1, \dots, N\}$, and for $y = N^{o(1)}$ let $S_y(f(n))$ be the y -smooth part and $R_y = f(n)/S_y(f(n))$ the residual cofactor. Then: (i) the small-prime Kubilius field generated by the valuations $v_p(f(n))$ for $p \leq y$ is asymptotically independent of the Frobenius-marked PD(1) process of R_y , provided the latter is normalized by $\log R_y$ and not by $\log |f(n)|$. (ii) [strong converse layer] the only leading coupling between the two scales is the removed logarithmic mass together with the explicit resultant data of f , so no further interaction survives. Clause (ii) is separately falsifiable from (i): residual dependence transmitted through Frobenius, or through values unusually close to smooth numbers, refutes it alone.*

Small-prime divisibility consumes logarithmic mass and therefore mechanically depresses the raw largest-factor share, so unqualified independence of the small and large scales is false by mass conservation, and the substantive statement is that once the remaining mass is used as the denominator, scale invariance restores independence. A proof would be a multiscale factorization principle allowing local sieve information and global factor-shape information to be combined without double counting, which is exactly the step that heuristics for polynomial values usually

take without justification. The Poisson–Dirichlet side is the theory of Bharadwaj and Rodgers, and the corrected conditional independence with the residual-mass normalization is the claimed content.

Evidence. For $n^3 - 2$ the raw Pearson correlation between the number of prime factors at most 100, counted with multiplicity, and the largest-factor logarithmic share was -0.465 and fell to 0.104 , with Spearman 0.074 , after renormalizing by the logarithm of the residual cofactor, so the correction is empirically essential, though 0.104 sits about three descriptive standard errors from zero at roughly 1200 samples and is therefore consistent with a slowly vanishing finite- N coupling rather than with exact independence at this range.

Conjecture 23 (Coloured factor superposition with local-exclusion gluing). *Let $f = \prod_{i=1}^s g_i \in \mathbb{Z}[x]$ be squarefree with irreducible components g_i , sample n uniformly from $\{1, \dots, N\}$, and let $y = N^{o(1)}$. Let $\rho(I, p) = \#\{x \bmod p : p \mid g_i(x) \text{ for all } i \in I\}$ be the joint root-count tensor. Then: (i) conditioned on the full joint local state $(v_p(g_i(n)))_{i \leq s, p \leq y}$, and with each component normalized by its own residual logarithmic mass, the coloured Frobenius-marked macroscopic factor processes of the g_i are asymptotically independent PD(1) processes. (ii) before conditioning, their entire interaction is the explicit Gibbs–Euler factor assembled from the tensors $\rho(I, p)$, whose leading part is*

$$\mathcal{G}(f) = \prod_p \frac{\sum_{I \subseteq \{1, \dots, s\}} (-1)^{|I|} \frac{\rho(I, p)}{p}}{\prod_{i=1}^s \left(1 - \frac{\rho(\{i\}, p)}{p}\right)},$$

the numerator being the density of residues modulo p avoiding every component by inclusion and exclusion and the denominator the same density computed as if the components were independent, so that the primes dividing the resultant contribute hard shared atoms and the remaining primes contribute exclusion or attraction according to whether $\rho(I, p)$ falls below or above the independent product. [stated programme] The closed assembly of the full Gibbs weight from the tensors $\rho(I, p)$ at every $|I|$, of which $\mathcal{G}(f)$ is the two-point part, is the derivation route registered with this conjecture. (iii) [strong converse layer] no further long-range colour coupling survives. Clause (iii) is separately falsifiable from (i) and (ii), and an archimedean size correlation or a medium-prime coupling persisting after every fixed- y conditioning refutes it alone.

Algebraic components generate separate macroscopic masses, but the same residue n modulo p couples their small-prime divisibility even when the resultant is one, so a reducible polynomial is not a product of independent random integers and deleting only the resultant primes is not enough. The statement supplies the correct functorial rule under multiplication of polynomials, predicting coloured largest factors, joint smoothness and common-cofactor risk without the false independence assumption that simpler heuristics use. The Poisson–Dirichlet background is again that of Bharadwaj and Rodgers, and the identification of the complete interaction as a Gibbs–Euler factor built from joint root-count tensors is what is claimed.

Evidence. For $g_1 = n^2 + 1$ and $g_2 = n^2 + n + 1$, whose resultant is one, no common prime factor occurred in 6000 samples and the largest-factor logarithmic shares had correlation -0.027 , about two descriptive standard errors from zero, but the computation does not condition on the joint local state, so clauses (i) and (iii) are untested and only the macroscopic null case predicted by clause (ii) at resultant one is observed.

Conjecture 24 (Saddle-point entanglement law for joint polynomial smoothness). *Let $f_1, \dots, f_s \in \mathbb{Z}[x]$ be pairwise coprime and primitive, sample n uniformly from $\{1, \dots, N\}$, and let the smoothness parameters $u_i = \log |f_i(n)| / \log y_i$ range over the admissible region, by which is meant a fixed compact subset of $\{(u_1, \dots, u_s) : u_i > 1 \text{ for every } i\}$ together with the restriction $y_i \geq (\log N)^{1+\varepsilon}$ for a fixed $\varepsilon > 0$, the range in which the saddle-point expansion for smooth values is available. For $J \subseteq \{1, \dots, s\}$ and a prime p let*

$$\mu_p^\sigma(J) = \frac{\rho(J, p) p^{-\sum_{i \in J} \sigma_i}}{\sum_{J' \subseteq \{1, \dots, s\}} \rho(J', p) p^{-\sum_{i \in J'} \sigma_i}}$$

be the tilted joint local measure attached to a vector $\sigma = (\sigma_1, \dots, \sigma_s)$, with $\rho(J, p)$ the joint root-count tensor of Section 1. The joint saddle equations are the s coupled equations

$$\sum_{p \leq y_i} \sum_{J \ni i} \mu_p^\sigma(J) \log p = \log |f_i(n)|, \quad 1 \leq i \leq s,$$

which determine σ and reduce for $s = 1$ to the classical saddle-point equation for smooth numbers. Then: (i) the joint density of n for which every $f_i(n)$ is y_i -smooth equals the product of the component Dickman probabilities multiplied by an Euler product of the tilted local root distributions μ_p^σ at the solution σ of the joint saddle equations, so the entanglement factor is a function of $(u_1, \dots, u_s)$ and not a fixed constant. (ii) [strong converse layer] after this tilt no additional long-range correlation between the components survives. Clause (ii) is separately falsifiable from (i), and a correlation persisting near the edge of the smoothness range, or an archimedean factor not captured by the Euler product, refutes it alone.

Conditioning on smoothness changes the effective probability that each small prime divides a value, so the local dependence between components must be computed under the tilted measure rather than under the unconditioned Euler product, and that is precisely why a fixed correction factor multiplying Dickman probabilities is too crude. A proof would give a coherent local-to-global theory of simultaneous smooth polynomial values and would correct a common heuristic error, linking saddle-point analysis, resultants and multivariate sieves. The single-polynomial theory rests on Dickman and on the saddle-point method for smooth numbers, in the uniform form developed by Hildebrand and Tenenbaum, and the parameter-dependent entanglement factor is the claimed content.

Evidence. For $n^2 + 1$ and $n^2 + n + 1$ the ratio of the joint probability to the product of the marginals was 0.674, 0.893 and 0.952 as the largest-factor threshold rose from 0.35 to 0.55, and the monotone movement of that ratio is what rejects a single fixed correction at this range, though the thresholds are largest-factor shares rather than genuine smoothness parameters and the sample is 6000 values.

Conjecture 25 (Buchstab–Bateman–Horn sieve-to-prime bridge). *Let $f \in \mathbb{Z}[x]$ be primitive, irreducible and admissible, sample n uniformly from $\{1, \dots, N\}$, and put $u = \log |f(n)| / \log y$ and $V_f(y) = \prod_{p \leq y} (1 - \rho_f(p)/p)$. Then: (i) uniformly for u in compact subsets of $(1, \infty)$, the density of y -rough values, those with no prime factor at most y , is $V_f(y) e^{\gamma \omega(u)} (1 + o(1))$. (ii) conditional on y -roughness, the probability that $f(n)$ is prime is*

$$\frac{C_f / \log |f(n)|}{V_f(y) e^{\gamma \omega(u)}} (1 + o(1)),$$

with C_f the Bateman–Horn constant of f . (iii) [elementary endpoint] for $1 < u \leq 2$ a y -rough value below y^2 has no admissible nontrivial factorization and is therefore prime unless it is a unit, so the conditional probability in (ii) equals one exactly on that range, the values being positive on the range considered and the unit case therefore not arising. The conditional probabilities of (ii) are consistent by construction as y increases, each being the ratio of the same Bateman–Horn density to the roughness density at that scale, and no martingale property is asserted.

The finite sieve accounts for the local factors up to y , the Buchstab evolution accounts for the number of remaining factors, and Bateman–Horn supplies the terminal one-factor density, so the parity barrier appears as a boundary condition at $u \leq 2$ rather than as an unexplained constant. A proof would reframe Bateman–Horn as a scale-by-scale law and would provide a concrete interface between sieve theory and prime-producing polynomials, predicting rough values, semiprime transitions and primality in a single equation. Bateman and Horn predict the terminal density, Buchstab theory predicts the finite-scale roughness, and the uniform conditional bridge between them, including the parity endpoint, is the distinctive statement.

Evidence. For $f(n) = n^2 + 1$ on $60,000 \leq n \leq 140,000$ and $y = 13, 31, 71, 151, 313$ the observed rough-survivor densities agreed with the finite- u prediction of clause (i) within 0.11 per cent and the observed conditional prime probabilities were uniformly 0.984 to 0.985 of the prediction of clause (ii) with no parameter fitted, and that uniform shortfall of about 1.5 per cent across five values of y is a systematic secondary term rather than sampling noise. The five rows are not five independent measurements: they share one nested sample of n , the rough survivors at a larger y being a subset of those at a smaller one, so the five agreements are strongly correlated and the consistency of the shortfall across the rows carries much less information than five independent confirmations would.

7 Reading the collection

The twenty-five statements are grouped by mechanism rather than by difficulty. Conjectures 1, 3, 6, 7, 11, 16, 18, 21, 24 and 25 are parent mechanisms, each supplying the objects that its programme then uses. Conjectures 2, 4, 14, 19 and 22 are independent structural extensions that introduce a cluster invariant, a valuation process or a conditional law not contained in the parent. Conjectures 8, 13 and 17 are phase boundaries. Conjecture 9 is a sharp anomaly statement, and Conjectures 5, 10, 12, 15, 20 and 23 are converse classifications or functorial laws.

Eleven conjectures carry a strong converse layer that is stated separately from its constructive companion and can fail on its own. The eleven tags are of one kind and form one list: Conjecture 1(ii) on the absence of coefficients outside the overlap-connected expansion, Conjecture 2(iii) on the absence of a variance scale outside the coincidence matrices and the renormalized disjoint kernel, Conjecture 3(ii) on the absence of a fourth covariance source, Conjecture 5(ii) on the converse classification of rigid coefficient sequences, Conjecture 7(iv) on the absence of mixed additive-multiplicative terms, Conjecture 9(iii) on the rank-one form of the exceptional-zero deformation, Conjecture 11(iii) on the absence of a constraint beyond the relation subtorus, Conjecture 20(ii) on the necessity of a geometric correspondence, Conjecture 22(ii) on the absence of a further small-large coupling, Conjecture 23(iii) on the absence of further colour coupling, and Conjecture 24(ii) on the absence of a correlation surviving the saddle-point tilt. Refuting a converse layer identifies a missing invariant and leaves the constructive layer of that conjecture standing, which is why the two are never merged into one clause.

Two clauses are deductions rather than conjectures and are labelled as such in place: Conjecture 1(iv), which derives Gaussianity and Wick pairing from the connected cumulant hypothesis of Conjecture 1(iii), and Conjecture 17(i), which derives the subcritical total-variation bound from an effective Chebotarev estimate under a Riemann hypothesis for the preimage fields. Four clauses are elementary and are proved by the argument stated with them: Conjecture 11(i), Conjecture 15(i), Conjecture 25(iii) and the identification of q_p with a connecting map recorded in Section 1. The constants $A(t)$ of Conjecture 6, $\theta(H)$ of Conjecture 4, θ_{occ} of Conjecture 8, $\kappa(\Gamma)$ of Conjecture 13, δ_p of Conjecture 19 and the mark intensity of Conjecture 21 are all derived from local, overlap, Galois or sieve structure, and where a closed evaluation is not yet available, as for $A(t)$, the derivation route is registered as the stated programme of that conjecture.

Finally, the computational range available to the evidence notes is small against every asymptotic asserted above. The prime-pattern windows overlap and carry no error bars, the occupancy moduli span less than one decade of $\log q$, the finite-logarithm tests use fixed frequencies only, the arboreal tests reach level four over ten effective maps, and the factorization tests use samples of a few thousand values. Each evidence note says which clause it touches and which it leaves untested, and none of them is offered as support for an asymptotic claim.